

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150RHP10HP001Ammonia HP flare KOD7%.pvdb

Date of Analysis : Sep 16, 2026 11:39am

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 3000.000 mm ID	
Vessel Design Length, Tangent to Tangent	9275.00 mm
Distance of Bottom Tangent above Grade	2300.00 mm
Distance of Base above Grade	300.00 mm
Specified Datum Line Distance	2000.00 mm
Internal Design Temperature	85 °C
Internal Design Pressure	0.350 MPa
External Design Temperature	85 °C
Maximum Allowable Working Pressure	1.09 MPa
Shop Test Pressure	1.422 MPa
Field Test Pressure	1.422 MPa
Required Minimum Design Metal Temperature	-40.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N+IT
Head	SA-516 70	...	...	K02700	+N+IT
Skirt	SA-285 C	...	...	K02801	+N
Skirt	SA-516 70	...	...	K02700	+N+IT
Nozzle	SA-350 LF1	1	...	K03009	+IT
Nozzle Flg	SA-350 LF1	1	...	K03009	+IT
Basering	SA-285 C	...	...	K02801	+N
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-2	...	...	...	1.5000	N/A	No
Skirt-1	...	...	...	1.5000	N/A	No
Btm_head	0.405	0.00	1.090	1.5000	No	No
Shell-1	0.399	0.00	1.090	1.5000	N/A	No
Shell-2	0.379	0.00	1.110	1.5000	N/A	No
Shell-3	0.359	0.00	1.130	1.5000	N/A	No
Shell-4	0.353	0.00	1.680	1.5000	N/A	No
Top_head	0.350	0.00	1.520	1.5000	No	No

Liquid Level: 8225.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.680

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
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Skirt	-1000.0	1000.0	14.0	14.0	...	...	0.70	0.70
Skirt	0.0	1000.0	14.0	14.0	...	...	0.70	0.70
Ellipse	50.0	50.0	18.0	14.0	5.9	...	1.00	1.00
Cylinder	3050.0	3000.0	14.0	14.0	5.9	...	1.00	1.00
Cylinder	6050.0	3000.0	14.0	14.0	5.6	...	1.00	1.00
Cylinder	7050.0	1000.0	14.0	14.0	5.4	...	1.00	1.00
Cylinder	9225.0	2175.0	20.0	20.0	5.3	...	1.00	1.00
Ellipse	9275.0	50.0	22.0	18.0	5.3	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
B2_3"	WRC-107/537	0.899	n/a	Passed
A2_3"	WRC-297	0.957	Shell	Passed
A3_2"	WRC-297	0.964	Shell	Passed
A1_18"	WRC-297	0.940	Shell	Passed
B1_18"	WRC-107/537	0.649	n/a	Passed

Wind/Earthquake Shear, Bending:

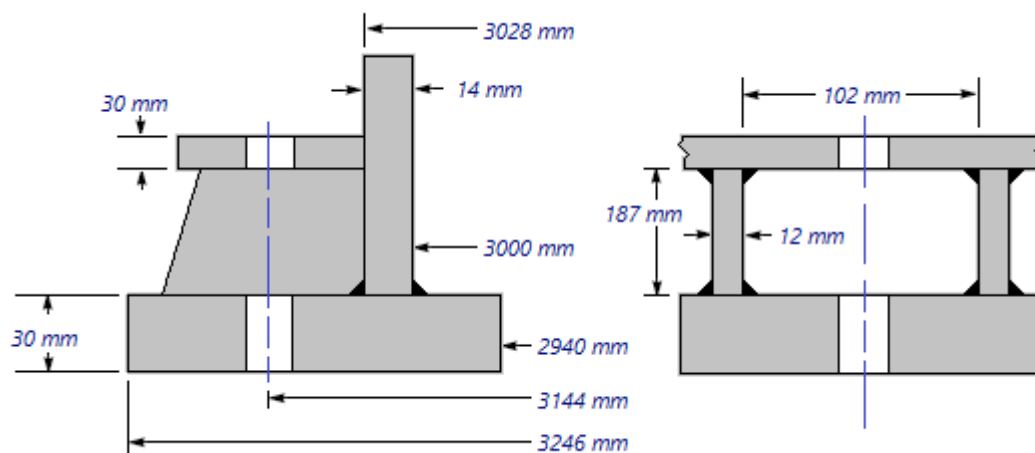
From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	500	97644.8	159956	736661824	1279528064
20	30	1500	94551.3	159659	640524800	1119656064
30	40	2025	91457.8	159110	547482560	960207040
40	50	3550	91303.2	156175	542911680	952321728
50	60	6550	70322.7	130254	300374560	522504352
60	70	8550	43085.4	84977.6	130193424	199525920
70	80	10137.5	39615.3	68120.4	88826312	122945864
80	90	11250	31817.6	33979.9	11111557	11866723

Abs Max of the all of the Stress Ratio's : 0.3021

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	30.0000	mm
Inside Diameter of Basing	2940.0000	mm
Outside Diameter of Basing	3246.0000	mm
Nominal Diameter of Bolts	30.0000	mm
Diameter of Bolt Circle	3144.0000	mm
Number of Bolts	12	
Thickness of Gusset Plates	12.0000	mm
Average Width of Gusset Plates	109.0000	mm
Height of Gussets	187.0000	mm
Distance between Gussets	102.0000	mm
Thickness of Top Plate or Ring	30.0000	mm
Radial Width of the Top Plate	109.0000	mm
Number of Gussets	24	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	97645.	N
Total Earthquake Shear on Support	159956.	N
Wind Moment on Support	739885824.	N-mm
Earthquake Moment on Support	1282752000.	N-mm
Wind Load Case Test Moment on Support	246322448.	N-mm
Earthquake Load Case Test Moment on Support	3224042.	N-mm
Test Case Wind Shear on Support	32223.	N
Maximum Initial Bolt Load	77278.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	20059.9	kgm
Shop Test - Fabricated + water (full)	93008.3	kgm
Shipping - Fab. + rem. intls.+ shipping app.	31289.7	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	32189.7	kgm
Empty - Fab. + intls. + details + wghts.	32189.7	kgm
Operating - Empty + operating liquid (no ca)	75055.4	kgm
Field Test - Empty weight + test medium (full)	104238.1	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
B2_3"	1.09	...	...	OK	6.30	OK	Passed
L2_2"	1.09	...	...	OK	4.92	OK	No Calc[*]
L4_2"	1.09	...	...	OK	4.92	OK	No Calc[*]
L6_2"	1.09	...	...	OK	4.92	OK	No Calc[*]
L8_2"	1.09	...	...	OK	6.30	OK	No Calc[*]
A2_3"	1.13	...	...	OK	6.30	OK	Passed
A3_2"	1.13	...	...	OK	4.92	OK	No Calc[*]
A1_18"	1.23	...	...	OK	9.83	OK	Passed
M1_24"	1.18	...	...	...		OK	Passed
L1_2"	1.68	...	...	OK	4.92	OK	No Calc[*]
L3_2"	1.68	...	...	OK	4.92	OK	No Calc[*]
L5_2'	1.68	...	...	OK	4.92	OK	No Calc[*]
L7_2"	1.68	...	...	OK	4.92	OK	No Calc[*]
B1_18"	1.22	...	...	OK	9.83	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1.09 Nozzle : L82"
 Minimum MAWP Shells/Flanges : 1.09 Element : Shell-1
 Minimum MAPnc Shells/Flanges : 1.28 Element : Shell-3

Computed Vessel M.A.W.P. : 1.09 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	B2_3"	0.000	0.000	144.862
40	L2_2"	300.000	315.000	110.034
40	L4_2"	300.000	0.000	110.034
40	L6_2"	300.000	45.000	110.034
40	L8_2"	300.000	90.000	107.800
60	A2_3"	6550.000	180.000	144.862
60	A3_2"	6050.000	0.000	96.068
70	A1_18"	8075.000	180.000	920.400
70	M1_24"	8250.000	0.000	1153.200
70	L1_2"	7500.000	315.000	122.034
70	L3_2"	7500.000	0.000	122.034
70	L5_2'	7500.000	45.000	122.034
70	L7_2"	7500.000	90.000	122.034
80	B1_18"	0.000	0.000	920.400

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
L2_2"	2.000 in	160	LWN	2.375	8.738	...	...	214.60	150
L4_2"	2.000 in	160	LWN	2.375	8.738	...	...	214.60	150
L6_2"	2.000 in	160	LWN	2.375	8.738	...	...	214.60	150
L8_2"	2.000 in Actual	LWN	3.260	16.000	...	...	214.97	150	
A3_2"	2.000 in	160	LWN	2.375	8.738	...	...	214.49	600
L1_2"	2.000 in	160	LWN	2.375	8.738	...	...	220.60	150
L3_2"	2.000 in	160	LWN	2.375	8.738	...	...	220.60	150
L5_2'	2.000 in	160	LWN	2.375	8.738	...	...	220.60	150
L7_2"	2.000 in	160	LWN	2.375	8.738	...	...	220.60	150
B2_3"	3.000 in	160	WNF	3.500	11.125	...	...	214.55	600
A2_3"	3.000 in	160	WNF	3.500	11.125	...	...	215.00	600
A1_18"	18.000 in Actual	WNF	18.945	12.000	...	...	297.25	150	
B1_18"	18.000 in Actual	WNF	18.945	12.000	...	...	282.75	150	
M1_24"	24.000 in Actual	WNF	24.000	18.000	...	...	351.29	150	

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grv Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grv Weld	Inside Weld
L2_2"	SA-350 LF1	12.000	10.000	...	...	...
L4_2"	SA-350 LF1	12.000	10.000	...	...	...
L6_2"	SA-350 LF1	12.000	10.000	...	...	...
L8_2"	SA-350 LF1	12.000	10.000	...	...	...
A3_2"	SA-350 LF1	12.000	10.000	...	...	...
L1_2"	SA-350 LF1	12.000	10.000	...	...	...
L3_2"	SA-350 LF1	12.000	10.000	...	...	...
L5_2'	SA-350 LF1	12.000	10.000	...	...	...
L7_2"	SA-350 LF1	12.000	10.000	...	...	...
B2_3"	SA-350 LF1	12.000	10.000	...	...	...
A2_3"	SA-350 LF1	12.000	10.000	...	...	...
A1_18"	SA-350 LF1	18.000	10.000	...	...	...
B1_18"	SA-350 LF1	12.000	10.000	...	...	...
M1_24"	SA-350 LF1	12.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
L2_2"	300.000	315.0	200.00	0.00	Shell-1
L4_2"	300.000	0.0	200.00	0.00	Shell-1
L6_2"	300.000	45.0	200.00	0.00	Shell-1
L8_2"	300.000	90.0	200.00	0.00	Shell-1
A3_2"	6050.000	0.0	200.00	0.00	Shell-3
L1_2"	7500.000	315.0	200.00	0.00	Shell-4
L3_2"	7500.000	0.0	200.00	0.00	Shell-4
L5_2'	7500.000	45.0	200.00	0.00	Shell-4
L7_2"	7500.000	90.0	200.00	0.00	Shell-4



B2_3"	...	0.0	200.00	0.00	Btm_head
A2_3"	6550.000	180.0	200.00	0.00	Shell-3
A1_18"	8075.000	180.0	250.00	0.00	Shell-4
B1_18"	...	0.0	250.00	0.00	Top_head
M1_24"	8250.000	0.0	300.00	0.00	Shell-4

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 18.0mm THK X 3000.0mm ID X 50.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 22.0mm THK X 3000.0mm ID X 50.0mm	SA-516 70
2	PLATFORM: 1100mm X 1500mm WIDE	
1	LADDER: 3000.0mm	...
1	LADDER: 3900.0mm	...
1	PLATFORM: 1100mm X 5000mm WIDE	
1	LADDER: 5000.0mm	...
1	CLASS 150 GR 1.4, 24.0" BLIND FLANGE(S)	SA-350 LF1
1	BASERING: 3246.0mm OD X 2940.0 ID X 30.0mm	SA-285 C
12	BASERING BOLTS: 30.0mm DIA	SA-36
12	BASERING BOLT NUTS: 30.0mm DIA	...
1	BASERING TOP RING: 3246.0mm OD X 3028.0 ID X 30.0mm	...
24	BASERING GUSSET PLATES: 109.0mm X 109.0mm X 187.0mm X 12.0mm	...
3	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3048.0mm X 1000.0mm	SA-285 C
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 280.8mm X 1000.0mm	SA-285 C
3	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3048.0mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 280.8mm X 1000.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 14.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 280.8mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 6376.7mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 14.0mm THK X 3048.0mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 6376.7mm X 1000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 20.0mm THK X 3048.0mm X 2175.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 6376.7mm X 2175.0mm	SA-516 70
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Btmhead	[10]	!	-40	-40		0.999	14.000	1.00	No
Btmhead	[7]	!	-40	-53		0.761	18.000	1.00	No
Shell-1	[8]	!	-40	-40		1.000	14.000	1.00	No
Shell-2	[8]	!	-40	-41		0.982	14.000	1.00	No
Shell-3	[8]	!	-40	-42		0.965	14.000	1.00	No
Shell-4	[8]	!	-40	-60		0.648	20.000	1.00	No
Tophead	[10]	!	-40	-56		0.721	18.000	1.00	No
Tophead	[7]	!	-40	-64		0.583	22.000	1.00	No
B23"	[1]	!	-29	-29	-29	0.900	14.000	1.00	No
Nozzle Flg	[5]	!	-29	-104					
L22"	[1]	!	-29	-29	-29	0.999	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-34					
L42"	[1]	!	-29	-29	-29	0.999	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-34					
L62"	[1]	!	-29	-29	-29	0.999	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-34					
L82"	[1]	!	-29	-29	-29	0.999	14.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
A23"	[1]	!	-29	-29	-29	0.962	14.000	1.00	No
Nozzle Flg	[5]	!	-29	-104					
A32"	[1]	!	-29	-29	-29	0.965	14.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
A118"	[1]	!	-29	-29	-29	0.646	20.000	1.00	No
Nozzle Flg	[5]	!	-29	-47					
M124"	[1]	!	-29	-29	-29	0.646	20.000	1.00	No
Nozzle Flg	[5]	!	-29	-47					
L12"	[1]	!	-29	-29	-29	0.646	20.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
L32"	[1]	!	-29	-29	-29	0.646	20.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
L52"	[1]	!	-29	-29	-29	0.646	20.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
L72"	[1]	!	-29	-29	-29	0.646	20.000	1.00	No
Nozzle Flg	[5]	!	-40	-40					
B118"	[1]	!	-29	-29	-29	0.649	18.000	1.00	No
Nozzle Flg	[5]	!	-29	-47					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: -18 -29

Required Minimum Design Metal Temperature -40.0 °C

Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)



[20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
[21] - Body Flange Bolting Material
[22] - Nozzle Flange Bolting Material
[23] - Stiffening Ring to Shell Weld
[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 20 40 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	0.35	MPa
Design Internal Temperature	85.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-40.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW	
Load Case 17	NP+VO+OW	
Load Case 18	FS+BS+IP+OW	
Load Case 19	FS+BS+EP+OW	
Load Case 20	BL+IP+OW	
Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm
Distance Upwind of Crest	Lh	0 mm
Distance from Crest to the Vessel	x	0 mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm
Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	1.500	
Site Class	D	
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	2.500
Force Factor		0.000
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg
M.A.W.P + Static Head to Nozzle	Y	
Consider MAP New and Cold in Noz. Design	N	
Consider External Loads for Nozzle Des.	Y	
Use ASME VIII-1 Appendix 1-9	N	
Perform Blast Load Analysis	No	
Material Database Year	Current w/Addenda or Code Year	

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:



Element#	1/8
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt-2
Distance "FROM" to "TO"	1000 mm
Skirt Outside Diameter	3028 mm
Diameter of Skirt at Base	3028 mm
Skirt Thickness	14 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	85 °C
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.25
Material Name	SA-285 C +N
Allowable Stress, Ambient	108 MPa
Allowable Stress, Operating	108 MPa
Allowable Stress, Hydrotest	184.5 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	191.57 MPa
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	K02801
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	2/8
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt-1
Distance "FROM" to "TO"	1000 mm
Skirt Outside Diameter	3028 mm
Diameter of Skirt at Base	3028 mm
Skirt Thickness	14 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	85 °C
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.25
Material Name	SA-516 70 +N+IT
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	242 MPa
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	3/8
Element From Node	30
Element To Node	40
Element Type	Elliptical
Description	Btmhead
Distance "FROM" to "TO"	50 mm



Inside Diameter	3000	mm
Element Thickness	14	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	18	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.35	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.25	
Material Name	SA-516 70	+N+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	-750	mm
Height/Length of Liquid	800	mm
Liquid Density	0.0006789	kg/cm³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	B23"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	128.31	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1	+IT

Element#	4/8	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	3000	mm
Element Thickness	14	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.35	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.25	
Material Name	SA-516 70	+N+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	850	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	8063040	mm²
Platform Weight	38395	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm²
Platform Width	1500	mm



Platform Height	1100	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	2.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	3000	mm
Unit Weight of Ladder	0.49	N/mm
Platform Length (top head platform)	0	mm
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.0006789	kg/cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	L22"	
Dist. from "FROM" Node / Offset dist	250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	150	
Layout Angle	315.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	70.083	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	L42"	
Dist. from "FROM" Node / Offset dist	250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	70.083	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	L62"	
Dist. from "FROM" Node / Offset dist	250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	150	
Layout Angle	45.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	70.083	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	L82"	
Dist. from "FROM" Node / Offset dist	250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	108.01	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	



Element#	5/8
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	3000 mm
Inside Diameter	3000 mm
Element Thickness	14 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.25
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	2000 mm
Platform Start Angle (degrees)	180.0
Platform End Angle (degrees)	0.0
Platform Wind Area	8501950 mm ²
Platform Weight	20373 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1100 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	2.0
Ladder Layout Angle	180.0
Ladder Start Elevation	0 mm
Ladder End Elevation	3900 mm
Unit Weight of Ladder	0.49 N/mm
Platform Length (top head platform)	0 mm

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3000 mm
Liquid Density	0.0006789 kg/cm ³

Element From Node	50
Detail Type	Weight
Detail ID	Clipwt
Dist. from "FROM" Node / Offset dist	2500 mm
Miscellaneous Weight	8158.6 N
Offset from Element Centerline	0 mm

Element#	6/8
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	1000 mm
Inside Diameter	3000 mm
Element Thickness	14 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	85 °C



Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.25
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1000 mm
Liquid Density	0.0006789 kg/cm³

Element From Node	60
Detail Type	Nozzle
Detail ID	A23"
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	3 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	103.25 N
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT

Element From Node	60
Detail Type	Nozzle
Detail ID	A32"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	74.049 N
Grade of Attached Flange	GR 1.4
Nozzle Matl	SA-350 LF1 +IT

Element#	7/8
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	Shell-4
Distance "FROM" to "TO"	2175 mm
Inside Diameter	3000 mm
Element Thickness	20 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.25
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	425 mm
Liquid Density	0.0006789 kg/cm³



Element From Node 70
 Detail Type Nozzle
 Detail ID A118"
 Dist. from "FROM" Node / Offset dist 1025 mm
 Nozzle Diameter 18 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 1722 N
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-350 LF1 +IT

Element From Node 70
 Detail Type Nozzle
 Detail ID M124"
 Dist. from "FROM" Node / Offset dist 1200 mm
 Nozzle Diameter 24 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle 0.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 3977 N
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-350 LF1 +IT

Element From Node 70
 Detail Type Nozzle
 Detail ID L12"
 Dist. from "FROM" Node / Offset dist 450 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 150
 Layout Angle 315.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 70.083 N
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-350 LF1 +IT

Element From Node 70
 Detail Type Nozzle
 Detail ID L32"
 Dist. from "FROM" Node / Offset dist 450 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 150
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 71.884 N
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-350 LF1 +IT

Element From Node 70
 Detail Type Nozzle
 Detail ID L52"
 Dist. from "FROM" Node / Offset dist 450 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule 160
 Nozzle Class 150
 Layout Angle 45.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 71.884 N
 Grade of Attached Flange GR 1.4
 Nozzle Matl SA-350 LF1 +IT

Element From Node 70
 Detail Type Nozzle
 Detail ID L72"
 Dist. from "FROM" Node / Offset dist 450 mm



Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	71.884	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1	+IT

Element From Node	70	
Detail Type	Weight	
Detail ID	A1Pipewt	
Dist. from "FROM" Node / Offset dist	1025	mm
Miscellaneous Weight	24515	N
Offset from Element Centerline	2747	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	Lifting lug	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	5883.6	N
Offset from Element Centerline	0	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	Topdavit	
Dist. from "FROM" Node / Offset dist	500	mm
Miscellaneous Weight	5393.3	N
Offset from Element Centerline	1650	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	A1Halfopenpi	
Dist. from "FROM" Node / Offset dist	500	mm
Miscellaneous Weight	3432.1	N
Offset from Element Centerline	0	mm

Element#	8/8	
Element From Node	80	
Element To Node	90	
Element Type	Elliptical	
Description	Tophead	
Distance "FROM" to "TO"	50	mm
Inside Diameter	3000	mm
Element Thickness	18	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	22	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.35	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.25	
Material Name	SA-516 70	+N+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	1000	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	13438400	mm ²
Platform Weight	44147	N
Platform Railing Weight	0	N/mm



Platform Grating Weight	0.017	kg/cm ²
Platform Width	5000	mm
Platform Height	1100	mm
Platform Clearance or End Offset	2500	mm
Platform Force Coefficient	2.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	5000	mm
Unit Weight of Ladder	0.49	N/mm
Platform Length (top head platform)	5000	mm

Element From Node	80	
Detail Type	Nozzle	
Detail ID	B118"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	18	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1156.4	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	

Element From Node	80	
Detail Type	Weight	
Detail ID	B1pipewt	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	20102	N
Offset from Element Centerline	2747	mm

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Skirt-2	...	...	1000	...	1000
Skirt-1	...	...	2000	...	1000
Btm_head	...	...	2050	...	50
Shell-1	...	...	5050	...	3000
Shell-2	...	...	8050	...	3000
Shell-3	...	...	9050	...	1000
Shell-4	...	...	11225	...	2175
Top_head	...	...	11275	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-2	...	...	14	1.5	3028	...	1
Skirt-1	...	...	14	1.5	3028	...	2
Btm_head	0.4048	0.4048	18	1.5	3000	138	3
Shell-1	0.3995	0.3995	14	1.5	3000	138	4
Shell-2	0.3795	0.3795	14	1.5	3000	138	5
Shell-3	0.3595	0.3595	14	1.5	3000	138	6
Shell-4	0.3528	0.3528	20	1.5	3000	138	7
Top_head	0.35	0.35	22	1.5	3000	138	8

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.W.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-2	...	...	No Calc	No Calc	14	No Calc	1
Skirt-1	...	...	No Calc	No Calc	14	No Calc	2
Btm_head	0.35	0.35	1.09	1.29	14	5.9001	3
Shell-1	0.35	0.35	1.09	1.28	14	5.85429	4
Shell-2	0.35	0.35	1.11	1.28	14	5.63598	5
Shell-3	0.35	0.35	1.13	1.28	14	5.4177	6
Shell-4	0.35	0.35	1.68	1.83	20	5.34494	7
Top_head	0.35	0.35	1.52	1.65	18	5.30412	8

Minimum 1.09 1.28

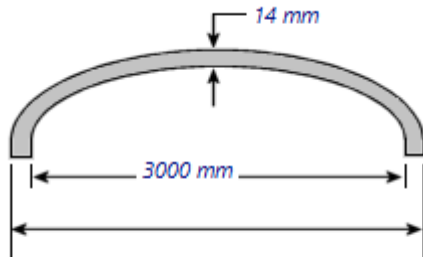
MAWP: 1.09 MPa, limited by: Shell-1.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Btmhead

Elliptical Head Design Information:					
Design Pressure	P	0.405 MPa	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		50.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress	48.6 MPa	
Spec. Min. Thk t,e, or sa		14.0000 mm	Spec. Nominal Thk.	18.000 mm	
Surface Area		10468307.0 mm ²	SG of Contents	0.680	
Element Volume		3888.4 ltr	Weight of Contents	25912.0 N	
Empty Weight		15561.0 N	Operating Weight	41473.0 N	
			Comp. All. Ope. Stress	103.5 MPa	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.4 \cdot 3003 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 0.4)$$

$$= 4.4001 + 1.5000 = 5.9001 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.055 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 12.5) / (1 \cdot 3003 + 0.2 \cdot 12.5)$$

$$= 1.15 - 0.055 = 1.09 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 14) / (1 \cdot 3000 + 0.2 \cdot 14)$$

$$= 1.29 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.4 \cdot (1 \cdot 3003 + 0.2 \cdot 12.5)) / (2 \cdot 1 \cdot 12.5)$$

= 48.603 MPa

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (0.4 \cdot 1502) / (138 \cdot 1 - 0.6 \cdot 0.4) + 1.5 \\ &= 5.912 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.050 MPa

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138 \cdot 1 \cdot 16.5) / (1502 + 0.6 \cdot 16.5) \\ &= 1.51 - 0.05 = 1.46 \text{ MPa} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6 \\ &= (2 + (3003 / (2 \cdot 752)))^2 / 6 \\ &= 0.998670 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 2.601 %

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

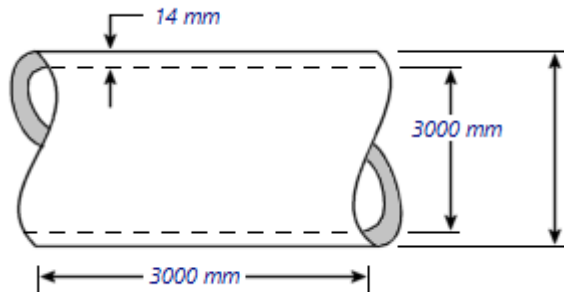
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	0.399 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		48.2 MPa
Spec. Min. Thk t _e or s _a		14.0000 mm	Spec. Nominal Thk.		14.000 mm
Surface Area		28538228.0 mm ²	SG of Contents		0.680
Element Volume		21209.6 ltr	Weight of Contents		141338.1 N
Empty Weight		30223.4 N	Operating Weight		171561.5 N
			Comp. All. Ope. Stress		103.5 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.4 \cdot 1502) / (138 \cdot 1 - 0.6 \cdot 0.4)$$

$$= 4.3543 + 1.5000 = 5.8543 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.049 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 12.5) / (1502 + 0.6 \cdot 12.5)$$

$$= 1.14 - 0.049 = 1.09 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (1500 + 0.6 \cdot 14)$$

$$= 1.28 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.4 \cdot (1502 + 0.6 \cdot 12.5)) / (1 \cdot 12.5)$$

$$= 48.227 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.464 %

Minimum Design Metal Temperature Results:



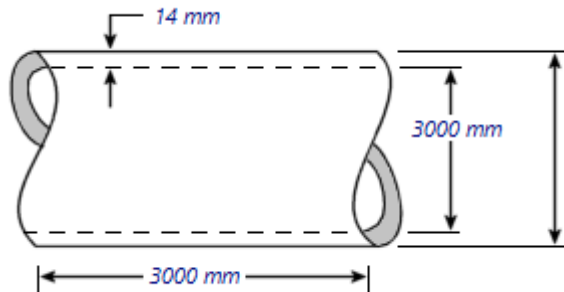
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	0.379 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		45.8 MPa
Spec. Min. Thk t _e or s _a		14.0000 mm	Spec. Nominal Thk.		14.000 mm
Surface Area		28538228.0 mm ²	SG of Contents		0.680
Element Volume		21209.6 ltr	Weight of Contents		141338.1 N
Empty Weight		30223.4 N	Operating Weight		171561.5 N
			Comp. All. Ope. Stress		103.5 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.38 \cdot 1502) / (138 \cdot 1 - 0.6 \cdot 0.38)$$

$$= 4.1360 + 1.5000 = 5.6360 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.029 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 12.5) / (1502 + 0.6 \cdot 12.5)$$

$$= 1.14 - 0.029 = 1.11 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (1500 + 0.6 \cdot 14)$$

$$= 1.28 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.38 \cdot (1502 + 0.6 \cdot 12.5)) / (1 \cdot 12.5)$$

$$= 45.813 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.464 %

Minimum Design Metal Temperature Results:



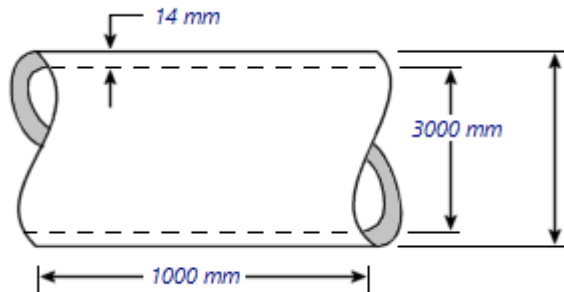
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	0.359 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		43.4 MPa
Spec. Min. Thk t _e , or s _a		14.0000 mm	Spec. Nominal Thk.		14.000 mm
Surface Area		9512743.0 mm ²	SG of Contents		0.680
Element Volume		7069.9 ltr	Weight of Contents		47112.7 N
Empty Weight		10074.4 N	Operating Weight		57187.1 N
			Comp. All. Ope. Stress		103.5 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.36 \cdot 1502) / (138 \cdot 1 - 0.6 \cdot 0.36)$$

$$= 3.9177 + 1.5000 = 5.4177 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.009 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 12.5) / (1502 + 0.6 \cdot 12.5)$$

$$= 1.14 - 0.0095 = 1.13 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (1500 + 0.6 \cdot 14)$$

$$= 1.28 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.36 \cdot (1502 + 0.6 \cdot 12.5)) / (1 \cdot 12.5)$$

$$= 43.399 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.464 %

Minimum Design Metal Temperature Results:



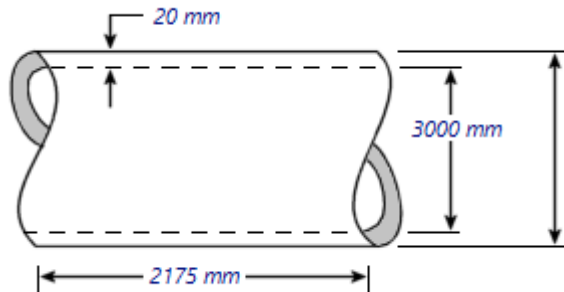
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	0.353 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		2175.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		28.8 MPa
Spec. Min. Thk t _e or s _a		20.0000 mm	Spec. Nominal Thk.		20.000 mm
Surface Area		20772214.0 mm ²	SG of Contents		0.680
Element Volume		15376.9 ltr	Weight of Contents		20022.9 N
Empty Weight		31365.1 N	Operating Weight		51388.0 N
			Comp. All. Ope. Stress		116.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.35 \cdot 1502) / (138 \cdot 1 - 0.6 \cdot 0.35)$$

$$= 3.8449 + 1.5000 = 5.3449 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.003 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 18.5) / (1502 + 0.6 \cdot 18.5)$$

$$= 1.69 - 0.0028 = 1.68 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 20) / (1500 + 0.6 \cdot 20)$$

$$= 1.83 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.35 \cdot (1502 + 0.6 \cdot 18.5)) / (1 \cdot 18.5)$$

$$= 28.848 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.662 %

Minimum Design Metal Temperature Results:



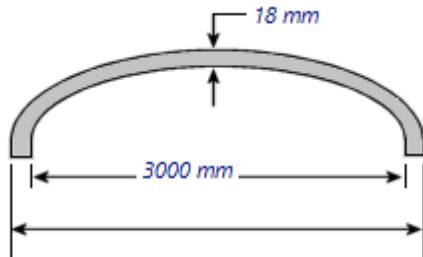
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Tophead

Elliptical Head Design Information:					
Design Pressure	P	0.350 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		31.8 MPa
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.		22.000 mm
Surface Area		10522290.0 mm ²	SG of Contents		0.000
Element Volume		3888.4 ltr	Weight of Contents		0.0 N
Empty Weight		19068.4 N	Operating Weight		19068.4 N
			Comp. All. Ope. Stress		113.0 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.35 \cdot 3003 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 0.35)$$

$$= 3.8041 + 1.5000 = 5.3041 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 16.5) / (1 \cdot 3003 + 0.2 \cdot 16.5)$$

$$= 1.52 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 18) / (1 \cdot 3000 + 0.2 \cdot 18)$$

$$= 1.65 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.35 \cdot (1 \cdot 3003 + 0.2 \cdot 16.5)) / (2 \cdot 1 \cdot 16.5)$$

$$= 31.843 \text{ MPa}$$



Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (0.35 \cdot 1502) / (138 \cdot 1 - 0.6 \cdot 0.35) + 1.5 \\ &= 5.314 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138 \cdot 1 \cdot 20.5) / (1502 + 0.6 \cdot 20.5) \\ &= 1.87 \text{ MPa} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (3003 / (2 \cdot 752))^2) / 6 \\ &= 0.998670 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 3.167 %

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm

Material treatment options: +N+IT

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.422 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.455 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.626 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.203 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.500 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.500 MPa
App 27-4	= M.A.W.P.	1.094 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 1.09 • 1
 = 1.422 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Btm_head	163.8	234.0	0.700	1.53
Shell-1	163.7	234.0	0.700	1.52
Shell-2	160.5	234.0	0.686	1.49
Shell-3	157.4	234.0	0.673	1.46
Shell-4	109.7	234.0	0.469	1.45
Top_head	119.3	234.0	0.510	1.43

Theoretical Maximum Allowable test pressure is : 2.076 MPa

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B2_3"	Nozzle	118.00	118.00	1.000
L2_2"	Nozzle	118.00	118.00	1.000
L4_2"	Nozzle	118.00	118.00	1.000
L6_2"	Nozzle	118.00	118.00	1.000
L8_2"	Nozzle	118.00	118.00	1.000
A2_3"	Nozzle	118.00	118.00	1.000
A3_2"	Nozzle	118.00	118.00	1.000
A1_18"	Nozzle	118.00	118.00	1.000
M1_24"	Nozzle	118.00	118.00	1.000
L1_2"	Nozzle	118.00	118.00	1.000
L3_2"	Nozzle	118.00	118.00	1.000
L5_2'	Nozzle	118.00	118.00	1.000
L7_2"	Nozzle	118.00	118.00	1.000
B1_18"	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Btm_head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Top_head	138.00	138.00	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
B2_3"	6.36	184.50	0.034
L2_2"	5.38	184.50	0.029
L4_2"	5.38	184.50	0.029
L6_2"	5.38	184.50	0.029
L8_2"	3.32	184.50	0.018
A2_3"	6.07	184.50	0.033
A3_2"	5.18	184.50	0.028
A1_18"	28.31	184.50	0.153
M1_24"	23.79	184.50	0.129
L1_2"	5.13	184.50	0.028
L3_2"	5.13	184.50	0.028
L5_2'	5.13	184.50	0.028
L7_2"	5.13	184.50	0.028
B1_18"	28.09	184.50	0.152

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.422 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.455 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.626 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.203 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.500 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.500 MPa
App 27-4	= M.A.W.P.	1.094 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 1.09 • 1
 = 1.422 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Btm_head	183.4	234.0	0.784	1.53
Shell-1	183.4	234.0	0.784	1.52
Shell-2	179.9	234.0	0.769	1.49
Shell-3	176.3	234.0	0.754	1.46
Shell-4	118.6	234.0	0.507	1.45
Top_head	130.1	234.0	0.556	1.43

Theoretical Maximum Allowable test pressure is : 1.841 MPa

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
B2_3"	Nozzle	118.00	118.00	1.000
L2_2"	Nozzle	118.00	118.00	1.000
L4_2"	Nozzle	118.00	118.00	1.000
L6_2"	Nozzle	118.00	118.00	1.000
L8_2"	Nozzle	118.00	118.00	1.000
A2_3"	Nozzle	118.00	118.00	1.000
A3_2"	Nozzle	118.00	118.00	1.000
A1_18"	Nozzle	118.00	118.00	1.000
M1_24"	Nozzle	118.00	118.00	1.000
L1_2"	Nozzle	118.00	118.00	1.000
L3_2"	Nozzle	118.00	118.00	1.000
L5_2"	Nozzle	118.00	118.00	1.000
L7_2"	Nozzle	118.00	118.00	1.000
B1_18"	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Btm_head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Top_head	138.00	138.00	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
B2_3"	6.36	184.50	0.034
L2_2"	5.38	184.50	0.029
L4_2"	5.38	184.50	0.029
L6_2"	5.38	184.50	0.029
L8_2"	3.32	184.50	0.018
A2_3"	6.07	184.50	0.033
A3_2"	5.18	184.50	0.028
A1_18"	28.31	184.50	0.153
M1_24"	23.79	184.50	0.129
L1_2"	5.13	184.50	0.028
L3_2"	5.13	184.50	0.028
L5_2'	5.13	184.50	0.028
L7_2"	5.13	184.50	0.028
B1_18"	28.09	184.50	0.152

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	2000	...	...	No Calc	No Calc
20	30	2000	...	...	No Calc	No Calc
30	40	No Calc	3028	12.5	0.00057335	57.3223
40	50	9775	3028	12.5	0.00010414	10.4113
50	60	9775	3028	12.5	0.00010414	10.4113
60	70	9775	3028	12.5	0.00010414	10.4113
70	80	9775	3040	18.5	0.00018766	18.7613
80	90	No Calc	3036	16.5	0.00075483	75.4661

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	14	No Calc	...	0.26
40	50	14	No Calc	...	0.057
50	60	14	No Calc	...	0.057
60	70	14	No Calc	...	0.057
70	80	20	No Calc	...	0.15
80	90	18	No Calc	...	0.46

Minimum

0.057

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	2000	No Calc	No Calc	No Calc
20	30	2000	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	9775	No Calc	No Calc	No Calc
50	60	9775	No Calc	No Calc	No Calc
60	70	9775	No Calc	No Calc	No Calc
70	80	9775	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	1668.38	...	1558.75	...	116.786
20	30	1027.37	...	917.756	...	71.9162
30	40	1586.89	3888.42	1454.65	3899.74	111.082
40	50	3082.13	21209.6	2753.26	21252	215.749
50	60	3082.13	21209.6	2753.26	21252	215.749
60	70	1027.37	7069.85	917.756	7084	71.9162
70	80	3198.56	15376.9	2960.13	15407.7	223.899
80	90	1944.57	3888.42	1811.98	3899.74	136.12
Total		16617	72642.72	15127	72795.16	1163

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
30	Liqd	2642.46	...	-375	...	Liquid: 10
30	Nozl	14.0004	...	-893	...	B2_3"
40	Plat	4189.52	0.21387E-04	850	...	PLAT:[1 OF 1]
40	Liqd	14413.4	...	1500	...	Liquid 20
40	Nozl	7.64726	1131.37	250	1131.37	L2_2"
40	Nozl	7.64726	1600	250	...	L4_2"
40	Nozl	7.64726	1131.37	250	-1131.37	L6_2"
40	Nozl	11.786	...	250	-1600	L8_2"
50	Plat	2223.08	...	2000	1537.63	PLAT:[1 OF 1]
50	Liqd	14413.4	...	1500	...	Liquid 30
50	Wght	832	...	2500	...	Clip_wt
60	Liqd	4804.48	...	500	...	Liquid 40
60	Nozl	11.2667	-1600	500	...	A2_3"
60	Nozl	8.08005	1600	...	...	A3_2"
70	Liqd	2041.9	...	212.5	...	Liquid 50
70	Nozl	187.897	-1625.01	1025	...	A1_18"
70	Nozl	433.957	1650	1200	...	M1_24"
70	Nozl	7.64726	1131.37	450	1131.37	L1_2"
70	Nozl	7.84375	1600	450	...	L3_2"
70	Nozl	7.84375	1131.37	450	-1131.37	L5_2'
70	Nozl	7.84375	...	450	-1600	L7_2"
70	Wght	2500	-2747	1025	...	A1_Pipe_wt
70	Wght	600	...	1000	...	Lifting lug
70	Wght	550	1650	500	...	Top_davit
70	Wght	350	...	500	...	A1_Half_open_pi
80	Plat	4817.17	...	1000	...	PLAT:[1 OF 1]
80	Nozl	126.179	...	916.006	...	B1_18"
80	Wght	2050	2747	...	...	B1_pipe_wt

Total Weight of Each Detail Type:

Platforms	11229.8
Liquid	38315.7
Nozzles	847.3
Weights	6882.0
Sum of the Detail Weights	57274.8 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	17780.6	17780.6	17780.6	17780.6	17780.6	17780.6
Nozzles	847.3	847.3	847.3	847.3	847.3	847.3
Wld Weights	1432.0	1432.0	1432.0	1432.0	1432.0	1432.0



Platforms	...	...	11229.8	11229.8	11229.8	11229.8
Empty Weights	...	...	...	900.0	900.0	...
Test Weights	...	350.0	...	...	...	...
Ope Weights	...	...	...	...	...	5450.0
Ope. Liquid	...	...	...	...	...	38315.7
Test Liquid	...	72598.4	...	...	...	...
Totals	20059.9	93008.3	31289.7	32189.7	32189.7	75055.4

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	20059.9 kgm
Shop Test Wt.	- Fabricated weight + water (full)	93008.3 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	31289.7 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	32189.7 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	32189.7 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	75055.4 kgm
Field Test Wt.	- Empty weight + test medium (full)	104238.1 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	28518.4 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	44094.2 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	15136.5 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	9512743	9424778
20	30	9512743	9424778
30	40	10468307	10227107
40	50	28538228	28274334
50	60	28538228	28274334
60	70	9512743	9424778
70	80	20772214	20498894
80	90	10522290	10227107

Total	127377488.0	125776112.0 mm ²
Total	127.4	125.8 m ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	1785.16	1785.16	1785.16	...	1785.16
20	30	1099.29	1099.29	1099.29	...	1099.29
30	40	1711.97	4354.43	5598.01	...	1711.97
40	50	7522.13	21935.6	28718.7	184993	7522.13
50	60	6352.97	20766.4	27549.6	33533168	6352.97
60	70	1118.64	5923.12	8184.17	...	1118.64
70	80	5575.5	10117.4	20393	125296	8075.5
80	90	7024.04	9074.04	10910.1	...	9074.04

Cumulative Vessel Weight

	Cumulative Ope	Cumulative	Cumulative
--	----------------	------------	------------



From	To	Wgt. No Liquid kgm	Oper. Wgt. kgm	Hydro. Wgt. kgm
10	20	36739.7	75055.4	104238
20	30	34954.5	73270.3	102453
30	40	33855.2	72171	101354
40	50	32143.3	67816.5	95755.6
50	60	24621.1	45881	67036.9
60	70	18268.2	25114.6	39487.3
70	80	17149.5	19191.4	31303.1
80	90	9074.04	9074.04	10910.1

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	33843460	33843460	33843460
20	30	33843460	33843460	33843460
30	40	33843460	33843460	33843460
40	50	33843460	33843460	33843460
50	60	33658464	33658464	33658464
60	70	125296	125296	125296
70	80	125296	125296	125296
80	90	...	...	...

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
B2_3"	7.94	8.52	85	600	GR 1.4	4.877	7.036	7.943	7.036
L2_2"	1.52	1.63	85	150	GR 1.4	...	...	...	...
L4_2"	1.52	1.63	85	150	GR 1.4	...	...	...	...
L6_2"	1.52	1.63	85	150	GR 1.4	...	...	...	...
L8_2"	1.52	1.63	85	150	GR 1.4	...	...	...	...
A2_3"	7.94	8.52	85	600	GR 1.4	4.878	7.036	7.943	7.036
A3_2"	7.94	8.52	85	600	GR 1.4	6.338	5.576	7.943	5.576
A1_18"	1.52	1.63	85	150	GR 1.4	1.869	1.475	1.520	0.411
M1_24"	1.52	1.63	85	150	GR 1.4	...	...	...	...
L1_2"	1.52	1.63	85	150	GR 1.4	...	...	...	...
L3_2"	1.52	1.63	85	150	GR 1.4	...	...	...	...
L5_2"	1.52	1.63	85	150	GR 1.4	...	...	...	...
L7_2"	1.52	1.63	85	150	GR 1.4	...	...	...	...
B1_18"	1.52	1.63	85	150	GR 1.4	1.558	1.520	1.520	0.723
Min Rating	1.475	1.630, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
B2_3"	111.125	0.50	1181217	4800	1.094	7.036	Pass
A2_3"	111.125	0.50	1181304	4800	1.094	7.036	Pass
A3_2"	77.851	0.50	525024	3200	1.094	5.576	Pass
A1_18"	500.888	1.20	42526884	28799	1.094	1.475	Pass
B1_18"	500.888	1.20	35739920	21600	1.094	1.520	Pass

Evaluating Flange on Nozzle [B23"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1181217) + 4 \cdot \text{abs}(4800)111.125 \leq \pi \cdot 111.125^3 \cdot ((7.943 - 1.094) + .500 \cdot 7.943)$$

$$= 0.21034E+08 \leq 0.46662E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B23" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(7.943, 0.500 \cdot 7.943 + 7.943 - 21033840.0 / (\pi \cdot 111.125^3))$$

$$= \min(7.943, 7.036)$$

$$= 7.036 \text{ MPa}$$

Evaluating Flange on Nozzle [A23"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1181304) + 4 \cdot \text{abs}(4800)111.125 \leq \pi \cdot 111.125^3 \cdot ((7.943 - 1.094) + .500 \cdot 7.943)$$

$$= 0.21035E+08 \leq 0.46662E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A23" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(7.943, 0.500 \cdot 7.943 + 7.943 - 21035196.0 / (\pi \cdot 111.125^3))$$

$$= \min(7.943, 7.036)$$

$$= 7.036 \text{ MPa}$$

Evaluating Flange on Nozzle [A32"]:



$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(525024) + 4 \cdot \text{abs}(3200) 77.851 \leq \pi \cdot 77.851^3 [(7.943 - 1.094) + .500 \cdot 7.943] \\
 &= 0.93972E+07 \leq 0.16044E+08 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle A32" [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(7.943, 0.500 \cdot 7.943 + 7.943 - 9397238.0 / (\pi \cdot 77.851^3)) \\
 &= \min(7.943, 5.576) \\
 &= 5.576 \text{ MPa}
 \end{aligned}$$

Evaluating Flange on Nozzle [A118"]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(42526884) + 4 \cdot \text{abs}(28799) 500.888 \leq \pi \cdot 500.888^3 [(1.520 - 1.094) + 1.200 \cdot 1.520] \\
 &= 0.73815E+09 \leq 0.88882E+09 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle A118" [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(1.520, 1.200 \cdot 1.520 + 1.520 - 738153216.0 / (\pi \cdot 500.888^3)) \\
 &= \min(1.520, 1.475) \\
 &= 1.475 \text{ MPa}
 \end{aligned}$$

Evaluating Flange on Nozzle [B118"]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(35739920) + 4 \cdot \text{abs}(21600) 500.888 \leq \pi \cdot 500.888^3 [(1.520 - 1.094) + 1.200 \cdot 1.520] \\
 &= 0.61513E+09 \leq 0.88882E+09 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle B118" [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(1.520, 1.200 \cdot 1.520 + 1.520 - 615133056.0 / (\pi \cdot 500.888^3)) \\
 &= \min(1.520, 1.787) \\
 &= 1.520 \text{ MPa}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 8.48172 Hz.

The Natural Frequency for the Vessel (Ope...) is 7.0231 Hz.

The Natural Frequency for the Vessel (Filled) is 5.78115 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.16 \cdot 0.94) / (1 + 1.7 \cdot 3.4 \cdot 0.16)))$
 $= \min(0.85, 0.9)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	7.023 Hz
Natural Frequency of Vessel (Empty)	8.482 Hz
Natural Frequency of Vessel (Test)	5.781 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.551
Structure Height to Diameter ratio	4.046
Height to top of Structure	12042.999 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$: $Zg = 589788$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 500 + 300 = 800$ mm
 $= 2.62$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (2.62 ft.) < 15 ft.
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15/1935)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf [1.59] kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.55 \cdot 40.7$$

$$= 632 \text{ lbs. [2812] N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-2	800.0	0.000	0.000	0.000	1.035	1.000	1.587
Skirt-1	1800.0	0.000	0.000	0.000	1.035	1.000	1.587
Btmhead	2325.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-1	3850.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-2	6850.0	0.000	0.000	0.000	1.110	1.000	1.703
Shell-3	8850.0	0.000	0.000	0.000	1.161	1.000	1.780
Shell-4	10437.5	0.000	0.000	0.000	1.195	1.000	1.832
Tophead	11874.1	0.000	0.000	0.000	1.222	1.000	1.874

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(104)$
 L - Total height of structure mm
 Lc - Total length of conical section(s) of vessel, mm
 tb - Uncorroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, km/hr
 Vc - Critical wind velocity, km/hr
 Vw - Maximum wind speed at top of structure, km/hr
 W - Total corroded weight of structure, N
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
 Z - Maximum amplitude of vibration at top of vessel, mm
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.31416E-05 (High); 0.31416E-05 < 0.39270E-05 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0/11275 = 0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[9.94/37^2] \cdot 10^4 = 72.7$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.39270E-05$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 720361 / (11275 \cdot 3033^2)$$

$$= 0.69466E-05$$

Since Vp is > 0.39270E-05 no further vibration analysis is required !



Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLAT:[1 OF 1]	8063040.00	3200.00	1.59	12796.83	2.00
PLAT:[1 OF 1]	8501950.00	7350.00	1.72	14641.22	2.00
PLAT:[1 OF 1]	13438400.00	12525.00	1.89	25360.52	2.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Clipwt	0.00	7850.00	1.74	0.00
A1Pipewt	0.00	10375.00	1.83	0.00
Lifting lug	0.00	10350.00	1.83	0.00
Topdavit	0.00	9850.00	1.81	0.00
A1Halfopen	0.00	9850.00	1.81	0.00
B1pipewt	0.00	11525.00	1.86	0.00

The Natural Frequency for the Vessel (Ope...) is 7.0231 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	800	3785	3784999	1.58714	3093.48
20	30	1800	3785	3784999	1.58714	3093.48
30	40	2325	3785	189250	1.58714	154.674
40	50	3850	3785	11354999	1.58714	20980.4
50	60	6850	3785	11354999	1.70275	27237.4
60	70	8850	3785	3784999	1.78032	3470.02
70	80	10437.5	3800	8265001	1.83215	7797.78
80	90	11874.1	3795	2478840	1.8737	31817.6

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		1.500	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	2.500	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(39.50.75)$
 = 0.315 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/7.02)
 = 0.142

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.31$, 0.14) per 12.8.2
 = 0.142

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.22 / (1.5 / 1.5)$
 = 0.217

Maximum value of C_s per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.093 / (0.14 (1.5 / 1.5))$
 = 0.651

Minimum value of C_s per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
 = $\max((0.044 \cdot 0.22 \cdot 1.5), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.22 \cdot 735993$
 = 159955.859 N

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.22 \cdot 735993$



= 31991.2 N

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 7.0231 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	500	17505.3	297.11	279.053	31991.2
20	30	1500	10779.6	548.875	515.516	31230.3
30	40	2025	42699.6	2935.12	1083.83	30761.7
40	50	3550	215100	25920.7	8348.49	28905.7
50	60	6550	203635	45276.4	13009.4	19556
60	70	8550	58082.1	16857.2	2990.16	10704.7
70	80	10137.5	99211.2	34140.5	17670.7	8180.04
80	90	11250	88980	33979.9	24704.6	3867.66



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	500	97644.8	159956	736661824	1279528064
20	30	1500	94551.3	159659	640524800	1119656064
30	40	2025	91457.8	159110	547482560	960207040
40	50	3550	91303.2	156175	542911680	952321728
50	60	6550	70322.7	130254	300374560	522504352
60	70	8550	43085.4	84977.6	130193424	199525920
70	80	10137.5	39615.3	68120.4	88826312	122945864
80	90	11250	31817.6	33979.9	11111557	11866723

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	97644.8	0.0031722	0.012406	0.24247E-04
20	30	94551.3	0.02728	0.047383	0.45162E-04
30	40	91457.8	0.048518	0.049666	0.46179E-04
40	50	91303.2	0.13986	0.2649	0.92787E-04
50	60	70322.7	0.41529	0.58303	0.00011612
60	70	43085.4	0.64166	0.70132	0.0001202
70	80	39615.3	0.83314	0.96639	0.00012284
80	90	31817.6	0.96946	0.97253	0.00012284

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	477.195	2982.47
20	30	477.195	2982.47
30	40	477.195	2982.47
40	50	477.195	2982.47
50	60	477.195	2982.47
60	70	477.195	2982.47
70	80	479.086	2994.29
80	90	478.456	2990.35

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 56.375 Actual deflection : 0.973 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	132563	118419	99426480	88905848
20	30	132563	118419	99426480	88905848
30	40	132563	118419	99426480	88905848
40	50	132563	118419	99426480	88905848
50	60	132563	118419	99426480	88905848
60	70	132563	118419	99426480	88905848
70	80	189753	175608	142326848	131847768
80	90	170663	156519	128006848	117513936

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Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	90.7184	154.977	-103.516	-103.516
20	30	115.92	196.56	-103.516	-103.516
30	40	165.6	280.8	-103.516	-103.516
40	50	165.597	280.795	-103.516	-103.516
50	60	165.597	280.795	-103.516	-103.516
60	70	165.597	280.795	-103.516	-103.516
70	80	165.597	280.795	-116.586	-116.586
80	90	165.597	280.795	-112.999	-112.999

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	20.951	...	85.104
40	50	20.951	...	85.104
50	60	20.951	...	85.104
60	70	20.951	...	85.104
70	80	14.1334	...	57.4104
80	90	15.855	...	64.4038

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-3.0426	-6.21571	-8.63247	0.38054	0.38054
20	30	-2.89476	-6.06787	-8.48463	0.38054	0.38054
30	40	-2.80372	-2.80372	-2.80372	0.38054	0.38054
40	50	-2.66194	-2.66194	-2.66194	0.38054	0.38054
50	60	-2.039	-2.039	-2.039	0.37846	0.37846
60	70	-1.51288	-1.51288	-1.51288	0.0014089	0.0014089
70	80	-0.95771	-0.95771	-0.95771	0.00095	0.00095
80	90	-0.56854	-0.56854	-0.56854	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	0.38054	8.28321	14.3873	2.73346	...
20	30	0.38054	7.20222	12.5897	2.37673	...
30	40	0.38054	6.15603	10.7968	2.03149	...
40	50	0.38054	6.10464	10.7081	2.01453	...
50	60	0.37846	3.37749	5.87517	1.11457	...
60	70	0.0014089	1.46393	2.24352	0.4831	...
70	80	0.00095	0.67349	0.93218	0.22225	...
80	90	...	0.094525	0.10095	0.031193	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	6.84593
20	30	...	...	...	6.07581
30	40	...	...	...	5.31017
40	50	...	...	...	5.27233
50	60	...	...	...	3.16153
60	70	...	...	...	1.41109
70	80	...	...	...	0.61875



80	90	...	...	...	0.073394
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Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	...	0.27018
20	30	...	0.26375
30	40	...	0.25979
40	50	...	0.24412
50	60	...	0.16516
60	70	...	0.090405
70	80	...	0.046585
80	90	...	0.024713

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...
80	90	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.62	90.72	-11.71	103.52	0.0620	0.1131
20	4.69	115.92	-10.48	103.52	0.0404	0.1012
30	3.73	165.60	-9.34	103.52	0.0225	0.0902
40	3.82	165.60	-9.15	103.52	0.0231	0.0884
50	1.72	165.60	-5.79	103.52	0.0104	0.0560
60		165.60	-2.98	103.52		0.0288
70		165.60	-1.63	116.59		0.0140
80		165.60	-0.66	113.00		0.0059

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.45	90.72	-10.00	103.52	0.0491	0.0966
20	3.83	115.92	-9.09	103.52	0.0330	0.0878
30	3.15	165.60	-8.23	103.52	0.0190	0.0795
40	3.24	165.60	-8.07	103.52	0.0195	0.0780
50	1.67	165.60	-5.41	103.52	0.0101	0.0523
60		165.60	-2.83	103.52		0.0274
70		165.60	-1.53	116.59		0.0131
80		165.60	-0.62	113.00		0.0055

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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Node	Stress	Stress	Stress	Stress	Ratio	Ratio
10	2.45	90.72	-14.88	103.52	0.0270	0.1437
20	1.51	115.92	-13.65	103.52	0.0131	0.1319
30	3.73	165.60	-9.34	103.52	0.0225	0.0902
40	3.82	165.60	-9.15	103.52	0.0231	0.0884
50	1.72	165.60	-5.79	103.52	0.0104	0.0560
60		165.60	-2.98	103.52		0.0288
70		165.60	-1.63	116.59		0.0140
80		165.60	-0.66	113.00		0.0059

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.82	90.72	-20.71	103.52	0.0972	0.2001
20	7.17	115.92	-18.77	103.52	0.0618	0.1814
30	8.63	165.60	-13.72	103.52	0.0521	0.1326
40	8.67	165.60	-13.51	103.52	0.0524	0.1305
50	4.38	165.60	-8.13	103.52	0.0264	0.0785
60	0.82	165.60	-3.67	103.52	0.0050	0.0354
70	0.02	165.60	-1.84	116.59	0.0001	0.0158
80		165.60	-0.64	113.00		0.0057

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-11.75	103.52		0.1135
20		196.56	-11.24	103.52		0.1086
30		280.80	-5.22	103.52		0.0504
40		280.79	-5.06	103.52		0.0489
50		280.79	-3.53	103.52		0.0341
60		280.79	-2.00	103.52		0.0193
70		280.79	-1.18	116.59		0.0101
80		280.79	-0.60	113.00		0.0053

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-9.01	103.52		0.0871
20		196.56	-8.87	103.52		0.0856
30		280.80	-3.18	103.52		0.0308
40		280.79	-3.04	103.52		0.0294
50		280.79	-2.42	103.52		0.0234
60		280.79	-1.51	103.52		0.0146
70		280.79	-0.96	116.59		0.0082
80		280.79	-0.57	113.00		0.0050

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.45	90.72	-14.88	103.52	0.0270	0.1437
20	1.51	115.92	-13.65	103.52	0.0131	0.1319
30	24.68	165.60		103.52	0.1491	
40	24.77	165.60		103.52	0.1496	
50	22.67	165.60		103.52	0.1369	
60	20.90	165.60		103.52	0.1262	
70	13.85	165.60		116.59	0.0836	
80	15.38	165.60		113.00	0.0929	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.82	90.72	-20.71	103.52	0.0972	0.2001
20	7.17	115.92	-18.77	103.52	0.0618	0.1814
30	29.58	165.60		103.52	0.1786	



40	29.62	165.60	103.52	0.1789
50	25.33	165.60	103.52	0.1530
60	21.77	165.60	103.52	0.1315
70	14.16	165.60	116.59	0.0855
80	15.41	165.60	113.00	0.0931

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.45	90.72	-14.88	103.52	0.0270	0.1437
20	1.51	115.92	-13.65	103.52	0.0131	0.1319
30	3.73	165.60	-9.34	103.52	0.0225	0.0902
40	3.82	165.60	-9.15	103.52	0.0231	0.0884
50	1.72	165.60	-5.79	103.52	0.0104	0.0560
60		165.60	-2.98	103.52		0.0288
70		165.60	-1.63	116.59		0.0140
80		165.60	-0.66	113.00		0.0059

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.82	90.72	-20.71	103.52	0.0972	0.2001
20	7.17	115.92	-18.77	103.52	0.0618	0.1814
30	8.63	165.60	-13.72	103.52	0.0521	0.1326
40	8.67	165.60	-13.51	103.52	0.0524	0.1305
50	4.38	165.60	-8.13	103.52	0.0264	0.0785
60	0.82	165.60	-3.67	103.52	0.0050	0.0354
70	0.02	165.60	-1.84	116.59	0.0001	0.0158
80		165.60	-0.64	113.00		0.0057

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-11.75	103.52		0.1135
20		196.56	-11.24	103.52		0.1086
30	84.71	280.80		103.52	0.3017	
40	84.84	280.79		103.52	0.3021	
50	84.56	280.79		103.52	0.3011	
60	84.08	280.79		103.52	0.2994	
70	56.68	280.79		116.59	0.2018	
80	63.87	280.79		113.00	0.2274	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-9.01	103.52		0.0871
20		196.56	-8.87	103.52		0.0856
30	82.68	280.80		103.52	0.2944	
40	82.82	280.79		103.52	0.2950	
50	83.44	280.79		103.52	0.2972	
60	83.59	280.79		103.52	0.2977	
70	56.45	280.79		116.59	0.2010	
80	63.84	280.79		113.00	0.2273	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-3.42	103.52		0.0331
20		115.92	-3.28	103.52		0.0316
30	18.53	165.60		103.52	0.1119	
40	18.67	165.60		103.52	0.1127	
50	19.29	165.60		103.52	0.1165	
60	19.44	165.60		103.52	0.1174	
70	13.18	165.60		116.59	0.0796	
80	15.29	165.60		113.00	0.0923	



Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-5.55	103.52		0.0536
20		115.92	-5.42	103.52		0.0524
30	18.45	165.60		103.52	0.1114	
40	18.57	165.60		103.52	0.1122	
50	19.13	165.60		103.52	0.1155	
60	19.60	165.60		103.52	0.1184	
70	13.25	165.60		116.59	0.0800	
80	15.33	165.60		113.00	0.0926	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-6.60	103.52		0.0637
20		115.92	-6.45	103.52		0.0623
30	18.53	165.60		103.52	0.1119	
40	18.67	165.60		103.52	0.1127	
50	19.29	165.60		103.52	0.1165	
60	19.44	165.60		103.52	0.1174	
70	13.18	165.60		116.59	0.0796	
80	15.29	165.60		113.00	0.0923	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-3.42	103.52		0.0331
20		115.92	-3.28	103.52		0.0316
30	18.53	165.60		103.52	0.1119	
40	18.67	165.60		103.52	0.1127	
50	19.29	165.60		103.52	0.1165	
60	19.44	165.60		103.52	0.1174	
70	13.18	165.60		116.59	0.0796	
80	15.29	165.60		113.00	0.0923	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-6.60	103.52		0.0637
20		115.92	-6.45	103.52		0.0623
30		165.60	-3.18	103.52		0.0308
40		165.60	-3.04	103.52		0.0294
50		165.60	-2.42	103.52		0.0234
60		165.60	-1.51	103.52		0.0146
70		165.60	-0.96	116.59		0.0082
80		165.60	-0.57	113.00		0.0050

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		75.60	-6.33	86.26		0.0733
20		96.60	-6.18	86.26		0.0717
30	18.79	138.00		86.26	0.1361	
40	18.91	138.00		86.26	0.1371	
50	19.46	138.00		86.26	0.1410	
60	19.53	138.00		86.26	0.1415	
70	13.22	138.00		97.15	0.0958	
80	15.31	138.00		94.17	0.1110	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10	75.60	-6.33	86.26	0.0733
20	96.60	-6.18	86.26	0.0717
30	138.00	-2.92	86.26	0.0339
40	138.00	-2.80	86.26	0.0324
50	138.00	-2.25	86.26	0.0261
60	138.00	-1.42	86.26	0.0165
70	138.00	-0.91	97.15	0.0094
80	138.00	-0.54	94.17	0.0058

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-6.60	103.52		0.0637
20		115.92	-6.45	103.52		0.0623
30	18.53	165.60		103.52	0.1119	
40	18.67	165.60		103.52	0.1127	
50	19.29	165.60		103.52	0.1165	
60	19.44	165.60		103.52	0.1174	
70	13.18	165.60		116.59	0.0796	
80	15.29	165.60		113.00	0.0923	

Absolute Maximum of the all of the Stress Ratio's 0.3021

Governing Element: Shell-1

Governing Load Case 11 : HP+HW+HI

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-2	...	-8.6	0.4	2.7	...	+ Tens
Skirt-2	...	-8.6	-0.4	-2.7	-11.7	- Comp
Skirt-1	...	-8.5	0.4	2.4	...	+ Tens
Skirt-1	...	-8.5	-0.4	-2.4	-11.2	- Comp
Btm_head	85.1	-2.8	0.4	2.0	84.7	+ Tens
Btm_head	85.1	-2.8	-0.4	-2.0	...	- Comp
Shell-1	85.1	-2.7	0.4	2.0	84.8	+ Tens
Shell-1	85.1	-2.7	-0.4	-2.0	...	- Comp
Shell-2	85.1	-2.0	0.4	1.1	84.6	+ Tens
Shell-2	85.1	-2.0	-0.4	-1.1	...	- Comp
Shell-3	85.1	-1.5	...	0.5	84.1	+ Tens
Shell-3	85.1	-1.5	...	-0.5	...	- Comp
Shell-4	57.4	-1.0	...	0.2	56.7	+ Tens
Shell-4	57.4	-1.0	...	-0.2	...	- Comp
Top_head	64.4	-0.6	...	...	63.9	+ Tens
Top_head	64.4	-0.6	...	...	...	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-1

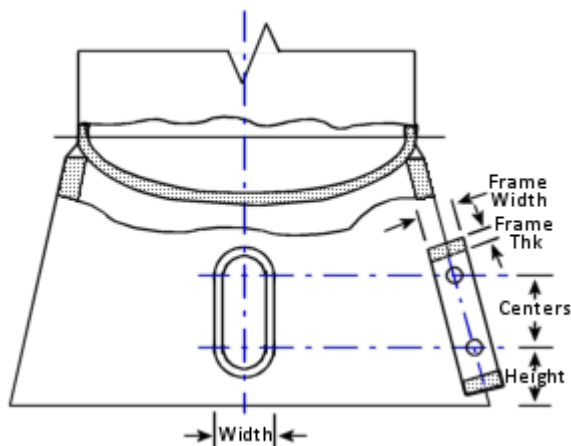
Total vessel (empty) weight	Fg	333834.844 N
Head Weight below section 2-2	DeltaFg	16787.576 N
Weight of vessel contents (liquid)	Ff	375723.906 N
Vessel Gross Weight	Fg + Ff	709558.688 N
Skirt Mean Diameter	Dz	3015.500 mm
Skirt Analysis Thickness (Corroded)	ez	12.500 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	0.000 N
Moment at section 1-1	M1	960207104.000 N-mm
Applied Moment - Bottom of Skirt	Mb	1119656064.000 N-mm
Weight at Bottom of Skirt	W	718487.938 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	3015.500 mm
Head Corroded Thickness	eb	12.500 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 752 \text{ mm} \\
 K &= D_i / (2 * h_i) = 3003 / (2 * 752) = 2 \\
 R &= D_i (0.44 * K + 0.02) = 3003 * (0.44 * 2 + 0.02) = 2700 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 3003 * ((0.5 / 2) - 0.08) = 511 \text{ mm}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	134600753152.000 mm ⁴
Moment of Inertia	Iyy =	134600753152.000 mm ⁴
Section Modulus	W =	89272680.000 mm ³
Section Metal Area	A =	118418.414 mm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{\min(I_{xx}, I_{yy}) / \text{Area}} \\
 &= \sqrt{\min(134.60\text{E}+09, 134.60\text{E}+09) / 118418} \\
 &= 1066 \text{ mm}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 1066/1000$$

$$= 1.066$$

Correct the Moment [M4]:

top of skirt is the critical end
 Controlling Moment [M4] = 960207040 N-mm

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -960207040/357090720 - 718488/118418$$

$$= -8.76 \text{ MPa}$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +960207040/357090720 - 718488/118418$$

$$= -3.38 \text{ MPa}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= 0 - 333835 - 375724 + 4 \cdot 960207040/3016$$

$$= 563620 \text{ N}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= 0 - 333835 - 375724 - 4 \cdot 960207040/3016$$

$$= -1982737 \text{ N}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (563620 + 16788 + 375724) / (\pi \cdot 3015 \cdot 12.5) + 0.35 \cdot 3015 / (4 \cdot 12.5)$$

$$= 29.18 \text{ MPa}$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-1982737 + 16788 + 375724) / (\pi \cdot 3015 \cdot 12.5) + 0.35 \cdot 3015 / (4 \cdot 12.5)$$

$$= 7.68 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (375724 + 16788) / (\pi \cdot 3015 \cdot 12.5) + 0.35 \cdot 3015 / (4 \cdot 12.5)$$

$$= 24.42 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= 563620 / (\pi \cdot 3016 \cdot 12.5)$$

$$= 4.76 \text{ MPa}$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -1982737 / (\pi \cdot 3016 \cdot 12.5)$$

$$= -16.74 \text{ MPa}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$

$$= 1 - 0.5(3015 + 12.5 - 3016 + 12.5) / (511 + 12.5)$$

$$= 0.98$$

$\gamma = \arccos(\cos(\gamma)) = \arccos(0.98) = 12.5 \text{ deg}$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(12.5^2 + 12.5^2 + 2 \cdot 12.5 \cdot 12.5 \cdot 0.98)}$$

$$= 12.4 \text{ mm}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 12.42 \cdot 563620.4 = 7.0059 \times 10^6 \text{ N-mm}$$

$$M_q = a \cdot F_{zq} = 12.42 \cdot -1982737.9 = -24.6459 \times 10^6 \text{ N-mm}$$

Correction factor [C]:

$$= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2$$

$$= 0.63 - 0.057 \left(\frac{12.5}{12.5} \right)^2$$

$$= 0.573$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.57 \cdot 6 \cdot 7005933 / (\pi \cdot 3015 \cdot 12.5^2)$$

$$= 16.27 \text{ MPa}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.57 \cdot 6 \cdot -24645896 / (\pi \cdot 3015 \cdot 12.5^2)$$

$$= -57.22 \text{ MPa}$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.57 \cdot 6 \cdot 7005933 / (\pi \cdot 3016 \cdot 12.5^2)$$

$$= 16.27 \text{ MPa}$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.57 \cdot 6 \cdot -24645896 / (\pi \cdot 3016 \cdot 12.5^2)$$

$$= -57.22 \text{ MPa}$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 0.49/119 = 0.0041$$

$$y = 125 \cdot eb/D_b = 125 \cdot 12.5/3015 = 0.52$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994$$

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.52 + 3.994 = 4.16$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b \left(\left(\frac{y}{y_A} \right) \cdot \alpha - 1 \right) / (4 \cdot eb)$$

$$= (0.35 + 0) \cdot 3015 \left(\left(\frac{12.5}{40} \right) \cdot 4.16 - 1 \right) / (4 \cdot 12.5)$$

$$= 6.45 \text{ MPa}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 29.2 - 16.3 + 6.45$$

$$= 19.37 \text{ MPa}$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 29.2 + 16.3 - 6.45$$

$$= 39.00 \text{ MPa}$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 7.68 - -57.2 + 6.45$$

$$= 71.35 \text{ MPa}$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 7.68 + -57.2 - 6.45$$

$$= -55.99 \text{ MPa}$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 24.4 + 16.3 + 6.45$$

$$= 47.14 \text{ MPa}$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 24.4 - 16.3 - 6.45$$

$$= 1.71 \text{ MPa}$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 24.4 + -57.2 + 6.45$$

$$= -26.35 \text{ MPa}$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 24.4 - -57.2 - 6.45$$

$$= 75.20 \text{ MPa}$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= 4.76 - 16.3$$

$$= -11.51 \text{ MPa}$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= 4.76 + 16.3$$

$$= 21.03 \text{ MPa}$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -16.7 - -57.2$$

$$= 40.48 \text{ MPa}$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -16.7 + -57.2$$

$$= -73.97 \text{ MPa}$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 138 (3 - (1 / 1.5)) \cdot (29.2 / 138)^2$$

$$= 409.89 \text{ MPa}$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 138 (3 - (1 / 1.5)) \cdot (7.68 / 138)^2$$

$$= 413.72 \text{ MPa}$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 138 (3 - (1 / 1.5)) \cdot (24.4 / 138)^2$$

$$= 411.12 \text{ MPa}$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$

$$= 138 (3 - (1 / 1.5)) \cdot (24.4 / 138)^2$$

$$= 411.12 \text{ MPa}$$

For $\sigma_{3p\dot{t}ot}$ & $\sigma_{3p\dot{o}tot}$, the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 138(3 - (1/1.5) \cdot (4.76/138)^2)$$

$$= 413.89 \text{ MPa}$$

For $\sigma_{3q\dot{t}ot}$ & $\sigma_{3q\dot{o}tot}$, the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 138(3 - (1/1.5) \cdot (-16.7/138)^2)$$

$$= 412.65 \text{ MPa}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	19.37	409.89	Pass	46
Stot1po	39.00	409.89	Pass	47
Stot1qi	71.35	413.72	Pass	48
Stot1qo	-55.99	413.72	Pass	49
Stot2pi	47.14	411.12	Pass	50
Stot2po	1.71	411.12	Pass	51
Stot2qi	-26.35	411.12	Pass	52
Stot2qo	75.20	411.12	Pass	53
Stot3pi	-11.51	413.89	Pass	54
Stot3po	21.03	413.89	Pass	55
Stot3qi	40.48	412.65	Pass	56
Stot3qo	-73.97	412.65	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	7721.639 mm
Center of Gravity of the Liquid	5477.155 mm
Center of Gravity of the Nozzles	9944.691 mm
Center of Gravity of the Added Weights (Operating)	10041.461 mm
Center of Gravity of the Added Weights (Empty)	8965.094 mm
Center of Gravity of Bare Shell New and Cold	6014.589 mm
Center of Gravity of Bare Shell Corroded	6034.842 mm
Vessel CG in the Operating Condition	6422.262 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	6927.312 mm
Vessel CG in the Test Condition	6726.478 mm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	3028.0000	mm
Skirt Thickness	STHK	14.0000	mm
Skirt Internal Corrosion Allowance	SCA	1.5000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		SA-285 C	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	30.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-285 C	
Basing Ope. All. Stress	BASOPE, f4	108.00	MPa
Basing Yield Stress		207.00	MPa
Inside Diameter of Basing	DI, D4	2940.0000	mm
Outside Diameter of Basing	DOU, D5	3246.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	30.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	114.00	MPa
Number of Bolts	RN, nb	12	
Diameter of Bolt Circle	DC, DBC	3144.0000	mm
Bolt Allowable Shear Stress		100.000	MPa
Ultimate Comp. Strength of Concrete	FPC	35.0	MPa
Allowable Comp. Strength of Concrete	FC	15.8	MPa
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	12.0000	mm
Width of Gussets at Top Plate	TWDT	109.0000	mm
Width of Gussets at Base Plate	BWDT	109.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	207.0	MPa
Height of Chair	h1	247.0000	mm
Distance between Gussets	RG, b6	102.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	51.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	30.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	109.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	36.0000	mm
External Corrosion Allowance	CA	1.5000	mm
Dead Weight of Vessel	DW	315652.0	N
Operating Weight of Vessel	ROW	735993.2	N
Test Weight of Vessel	TW	1022158.2	N
Earthquake Moment on Basing	EQMOM	1282752000.0	N-mm
Wind Moment on Basing	WIMOM	739885824.0	N-mm
Test Moment on Basing	TM	246322448.0	N-mm
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 1282752000/3144) - 704002) / 12$$

$$= 77278.0 \text{ N}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 739885824/3144) - 315652) / 12$$

$$= 52108.1 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/3144) - 735993) / 12$$

$$= -61332.8 \text{ N}$$

Bolt Load, Test Moment Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 246322448/3144) - 1022158) / 12$$

$$= -59074.8 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded)	Abss :	390.7817	mm ²
Area Available in all the Bolts	Abss * RN :	4689.3804	mm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.277	0.138	1.446	2.493	0.443	0.780	123319.4	827321.4	42768.0
4	0.208	0.242	1.242	2.644	0.457	0.776	110601.3	814603.3	36161.5
6	0.260	0.268	1.397	2.530	0.446	0.779	120106.9	824108.9	41033.6
8	0.264	0.266	1.409	2.521	0.445	0.779	120907.9	824909.9	41461.7
10	0.265	0.265	1.412	2.518	0.445	0.779	121108.4	825110.4	41569.3
12	0.265	0.265	1.413	2.518	0.445	0.779	121158.4	825160.4	41596.2
13	0.265	0.265	1.413	2.518	0.445	0.779	121133.4	825135.4	41582.8

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 121133 / (0.47 \cdot 3144 \cdot 2.52)$$

$$= 64.5 \text{ MPa, Should be } \leq 114 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 12 \cdot 391 / (\pi \cdot 3144)$$

$$= 0.47 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 2.36[(2 \cdot 0.27 \cdot 3144 + 153) / (2 \cdot 0.27 \cdot 3144)]$$

$$= 2.58 \text{ MPa, Should be } \leq 15.8 \text{ MPa}$$

Values for table 10.3, l = 109, b = 412, l/b = 0.26485333

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(2709, 13576) = 13576 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 13576 / 137)} + 1.5$$

$$= 25.9 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 109, b = 102, b/l = 0.93577987

As b/l (0.94) is less than 1, inverting b/l = 1.07 0.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l) / (\pi \cdot g))) + 1) - [(0.7 - g2)P / (4\pi)]$$

$$= 44545 / (4 \cdot \pi) [1.3 (\ln((2 \cdot 109 \cdot \sin(\pi \cdot 58/109)) / (\pi \cdot 25))) + 1] - [(0.7 - 0.13) \cdot 44545 / (4 \cdot \pi)]$$

$$= 6199.4922 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(6199) / 138) + 1.5}$$

$$= 17.9186 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 44545 / (124 \cdot 109) + 1.5$$

$$= 11 \text{ (not less than } 9.52 + 1.5 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 3144 / 12 = 823 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (44545 \cdot 58 / (823 \cdot 217 \cdot 162))^{2/3} \cdot 1514^{1/3} + \text{Ca}$$

$$= 4.04 + 1.5 = 5.54 \text{ mm}$$

Shear Stress in a Single Bolt [rb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 159956 / (391 \cdot 12)$$

$$= 34.1 \text{ MPa. Must be less than } 100.0 \text{ MPa.}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	25.9185	mm
Actual Basing Thickness as entered by user	30.0000	mm
Required Thickness of Chair Cap	17.9186	mm
Actual Top Ring Thickness as entered by user	30.0000	mm
Required Gusset thickness, + CA	11.0250	mm
Actual Gusset Thickness as entered by user	12.0000	mm
Required Thickness of Skirt for Local Stress	5.5375	mm
Given Thickness of Skirt	14.0000	mm
Required Gusset Height to meet local stress	9.8353	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 20.7 \cdot (14 - 1.5)$$

$$= 259 \text{ N/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 259 / [(0.4 \cdot 192) \cdot 2 \cdot 0.707]$$

$$= 2.39 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 44545 / (176 + 2 \cdot (187 + 30))$$

$$= 73 \text{ N/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 44545 \cdot 58 / (176 \cdot (217) + 0.6667 \cdot (217)^2)$$

$$= 37.1 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{73^2 + 37.1^2}$$

= 81.9 N/mm

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

= $W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
= $81.9 / [(0.4 \cdot 192) \cdot 2 \cdot 0.707]$
= 0.76 mm

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

= Bolt Load / (2 • TopWth)
= $44545 / (2 \cdot 109)$
= 204 N/mm

Weld Load [Wh]:

= Bolt Load • e / (2 • Hgt • TopWth)
= $44545 \cdot 58 / (2 \cdot 217 \cdot 109)$
= 54.6 N/mm

Resultant Weld Load [Wr]:

= $\sqrt{W_v^2 + W_h^2}$
= $\sqrt{204^2 + 54.6^2}$
= 212 N/mm

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

= $W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
= $212 / [(0.4 \cdot 192) \cdot 2 \cdot 0.707]$
= 1.95 mm

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	4.7625	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	1.9522	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: B23"

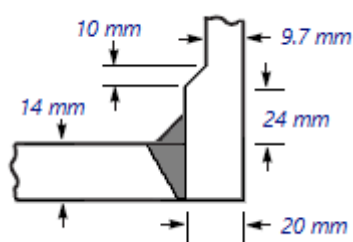
From: 30

Pressure for Reinforcement Calculations	P	1.1485	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	3000.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	14.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	24.0000	mm
Height of Beveled Transition	L'	10.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: B23"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.734 in.
 Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.15 \cdot 3003 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.15)$
 $= 11.2447 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 36.2 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 0.3545 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 10.3 / 0.5773$
 $= 17.7803 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

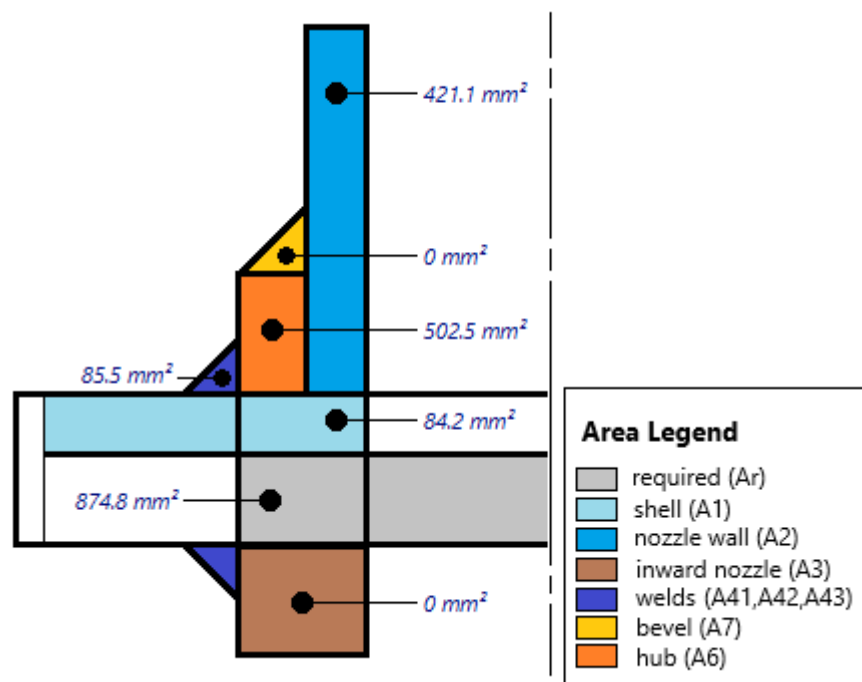
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	144.8618	mm
Parallel to Vessel Wall, opening length	d	72.4309	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	31.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	874.771	NA	NA
Area in Shell	A1	84.188	NA	NA
Area in Nozzle Wall	A2	421.116	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.506	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	502.460	NA	NA
TOTAL AREA AVAILABLE	Atot	1093.269	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (72.4 \cdot 11.2 \cdot 1 + 2 \cdot 18.5 \cdot 11.2 \cdot 1 \cdot (1 - 0.86))$
 $= 874.771 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 72.4(1 \cdot 12.5 - 1 \cdot 11.2) - 2 \cdot 18.5$
 $(1 \cdot 12.5 - 1 \cdot 11.2) \cdot (1 - 0.86)$
 $= 84.188 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 31.3)(8.23 - 0.35)0.86$
 $= 421.116 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 85.506 \text{ mm}^2$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(31.3, 200, 24)) \cdot (20 - 9.73) \cdot 0.86$$

$$= 502.460 \text{ mm}^2 \text{ Includes Bevel Area of } 81.137 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 1.8545 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.0000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 13.9894 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 13.9894 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 3.0000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 6.3000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[6.3, \max(14, 3)]$$

$$= 6.3000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1.85, 6.3)$$

$$= 6.3000 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 30.6,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 264.4 MPa	Passed
Occasional	: 2.3,	Allowable	: 156.9 MPa	Passed
Shear	: 22.3,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 1.15/8.52 = 0.135$$

Weld Size Calculations, Description: B23"

Intermediate Calc. for nozzle/shell Welds	T_{min}	16.5000 mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------

Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (875 - 84.2 + 2 \cdot 18.5 \cdot 0.86 \cdot (1 \cdot 12.5 - 11.2)) \cdot 138)$
 $= 114571.02 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (924 + 85.5 - 0 \cdot 0.86) \cdot 138$
 $= 139241.31 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (924 + 0 + 85.5 + (395)) \cdot 138$
 $= 193810.70 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (924 + 0 + 85.5 + 0 + (395)) \cdot 138$
 $= 193810.70 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 109 \cdot 10 \cdot 0.49 \cdot 118$
 $= 99379. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 45.5) \cdot (20 - 1.5) \cdot 0.7 \cdot 118$
 $= 218242. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 109 \cdot (12 - 0) \cdot 0.74 \cdot 138$
 $= 210627. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (99379 + 218242) = 317621 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (99379 + 0 + 210627 + 0) = 310006 \text{ N}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (99379 + 210627 + 0) = 310006 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 317620 N	, must exceed W = 114571 N	or W1 = 139241 N
Path 2-2 = 310006 N	, must exceed W = 114571 N	or W2 = 193810 N
Path 3-3 = 310006 N	, must exceed W = 114571 N	or W3 = 193810 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.09 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.



The Drop for this Nozzle is : 0.5523 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.5522 mm

Input Echo, WRC107/537 Item 1, Description: B23" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	5400.000	mm
Vessel Thickness	Tv	14.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	69.431	mm
Nozzle Thickness	Tn	20.000	mm
Nozzle Material		SA-350 LF1	
Nozzle UNS Number		K03009	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Design Internal Pressure	Dp	1.148	MPa
Include Pressure Thrust		No	

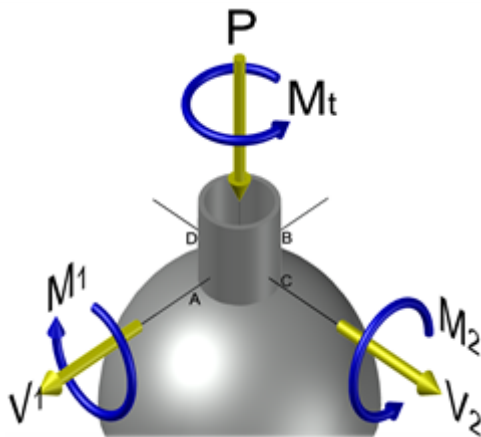
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-4799.8	N
Longitudinal Shear (SUS)	(V1) V1	4242.4	N
Circumferential Shear (SUS)	(Vc) V2	4242.4	N
Circumferential Moment (SUS)	(Mc) M1	835246.2	N-mm
Longitudinal Moment (SUS)	(Ml) M2	835246.2	N-mm
Torsional Moment (SUS)	Mt	1080379.9	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 109.431 + 2 \cdot 1.65 \cdot \sqrt{(2707.75 (14.0 - 1.5))}$$

$$= 716.549 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-4799.8	N
Circumferential Shear	(VC) V2	4242.4	N
Longitudinal Shear	(VL) V1	4242.4	N
Circumferential Moment	(MC) M1	835246.2	N-mm
Longitudinal Moment	(ML) M2	835246.2	N-mm
Torsional Moment	MT	1080379.9	N-mm

Dimensionless Param: U = 0.30 TAU = 5.00 (2.46) RHO = 0.68

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.10441	(A,B,C,D)
$M(x) / P$	SP 1	0.14034	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.17854	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.46690	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.17854	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.46690	(A,B,C,D)
$N(y) * T / P$	SP 1	0.16223	(A,B,C,D)
$M(y) / P$	SP 1	0.04885	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.05602	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.18035	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.05602	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.18035	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Rad. Bend.	P	25.9	-25.9	25.9	-25.9	25.9	-25.9	25.9	-25.9
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-5.2	-5.2	5.2	5.2
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-81.4	81.4	81.4	-81.4
Rad. Memb.	ML	-5.2	-5.2	5.2	5.2	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-81.4	81.4	81.4	-81.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-57.5	53.5	115.6	-98.8	-57.5	53.5	115.6	-98.8



Tang. Memb. P	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Tang. Bend. P	9.0	-9.0	9.0	-9.0	9.0	-9.0	9.0	-9.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Tang. Bend. MC	0.0	0.0	0.0	0.0	-31.4	31.4	31.4	-31.4
Tang. Memb. ML	-1.6	-1.6	1.6	1.6	0.0	0.0	0.0	0.0
Tang. Bend. ML	-31.4	31.4	31.4	-31.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-19.1	25.8	47.0	-33.8	-19.1	25.8	47.0	-33.8
Shear VC	2.0	2.0	-2.0	-2.0	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-2.0	-2.0	2.0	2.0
Shear MT	4.6	4.6	4.6	4.6	4.6	4.6	4.6	4.6
Tot. Shear	6.6	6.6	2.6	2.6	2.6	2.6	6.6	6.6
Str. Int.	58.6	55.0	115.7	99.0	57.7	53.8	116.3	99.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	124.1	124.1	124.1	124.1	124.1	124.1	124.1	124.1
Rad.	Pl (SUS)	-2.0	-2.0	8.4	8.4	-2.0	-2.0	8.4	8.4
Rad.	Q (SUS)	-55.5	55.5	107.2	-107.2	-55.5	55.5	107.2	-107.2
Long.	Pm (SUS)	124.1	124.1	124.1	124.1	124.1	124.1	124.1	124.1
Long.	Pl (SUS)	3.4	3.4	6.6	6.6	3.4	3.4	6.6	6.6
Long.	Q (SUS)	-22.4	22.4	40.4	-40.4	-22.4	22.4	40.4	-40.4
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	2.0	2.0	-2.0	-2.0	-2.0	-2.0	2.0	2.0
Shear	Q (SUS)	4.6	4.6	4.6	4.6	4.6	4.6	4.6	4.6
Pm (SUS)		124.1	124.1	124.1	124.1	124.1	124.1	124.1	124.1
Pm+Pl (SUS)		128.1	128.1	133.8	133.8	128.1	128.1	133.8	133.8
Pm+Pl+Q (Total)		106.1	179.1	239.8	90.4	105.2	177.9	240.4	90.9

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	124.10	138.00	Passed
Pm+Pl (SUS)	133.77	207.00	Passed
Pm+Pl+Q (TOTAL)	240.36	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: L22"

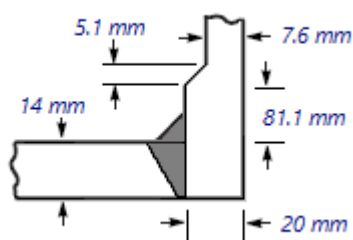
From: 40

Pressure for Reinforcement Calculations	P	1.1415	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		300.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		315.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	81.1239	mm
Height of Beveled Transition	L'	5.1239	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.14 \cdot 1502 / (138 \cdot 1 - 0.6 \cdot 1.14)$
 $= 12.4817 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.14 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 1.14)$
 $= 0.2337 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 18.5$, less than the hub height, use (e-2)
 $= 46.2500 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	110.0342	mm
Parallel to Vessel Wall	Rn+tn+t	55.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	31.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L22".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7337	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 13.9817	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.9817	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(14, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.73, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-34 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.14/1.63 = 0.700$

Weld Size Calculations, Description: L22"

Intermediate Calc. for nozzle/shell Welds Tmin 12.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6027 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.6027 mm

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Input, Nozzle Desc: L42"

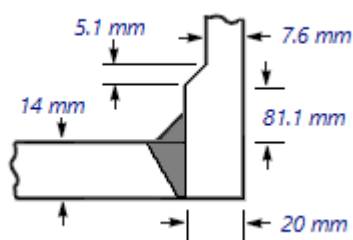
From: 40

Pressure for Reinforcement Calculations	P	1.1415	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		300.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	81.1239	mm
Height of Beveled Transition	L'	5.1239	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.14 \cdot 1502 / (138 \cdot 1 - 0.6 \cdot 1.14)$
 $= 12.4817 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.14 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 1.14)$
 $= 0.2337 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 18.5$, less than the hub height, use (e-2)
 $= 46.2500 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	110.0342	mm
Parallel to Vessel Wall	Rn+tn+t	55.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	31.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L42".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7337	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 13.9817	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.9817	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(14, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.73, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-34 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.14/1.63 = 0.700$

Weld Size Calculations, Description: L42"

Intermediate Calc. for nozzle/shell Welds Tmin 12.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6027 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.6027 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 55.02 mm

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Input, Nozzle Desc: L62"

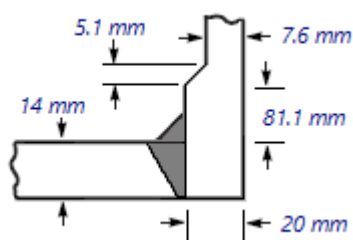
From: 40

Pressure for Reinforcement Calculations	P	1.1415	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		300.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	81.1239	mm
Height of Beveled Transition	L'	5.1239	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L62"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.14 \cdot 1502 / (138 \cdot 1 - 0.6 \cdot 1.14)$
 $= 12.4817 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.14 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 1.14)$
 $= 0.2337 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 18.5$, less than the hub height, use (e-2)
 $= 46.2500 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	110.0342	mm
Parallel to Vessel Wall	Rn+tn+t	55.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	31.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L62".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7337	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 13.9817	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.9817	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(14, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.73, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-34 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.14/1.63 = 0.700$

Weld Size Calculations, Description: L62"

Intermediate Calc. for nozzle/shell Welds Tmin 12.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6027 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.6027 mm

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Input, Nozzle Desc: L82"

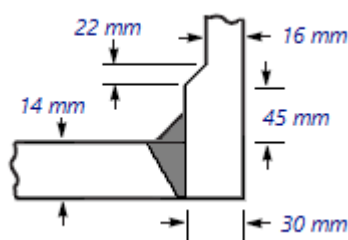
From: 40

Pressure for Reinforcement Calculations	P	1.1415	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		300.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	45.0000	mm
Height of Beveled Transition	L'	22.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L82"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.14 \cdot 1502 / (138 \cdot 1 - 0.6 \cdot 1.14)$
 $= 12.4817 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([1.14 / (118 \cdot 1)] - 1))$
 $= 0.2615 \text{ mm}$

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 12.5 / 0.5773$
 $= 21.6506 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.8000	mm
Parallel to Vessel Wall	Rn+tn+t	53.9000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	31.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L82".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7615	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 13.9817	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.9817	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(14, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.76, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification -29 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: L8_2"

Intermediate Calc. for nozzle/shell Welds Tmin 12.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9687 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.9688 mm

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Input, Nozzle Desc: A23"

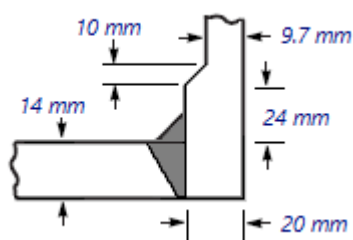
From: 60

Pressure for Reinforcement Calculations	P	1.0998	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6550.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	24.0000	mm
Height of Beveled Transition	L'	10.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: A23"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.734 in.
 Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1502 / (138 \cdot 1 - 0.6 \cdot 1.1)$
 $= 12.0239$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 36.2 / (118 \cdot 1 - 0.6 \cdot 1.1)$
 $= 0.3394$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 10.3 / 0.5773$
 $= 17.7803$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

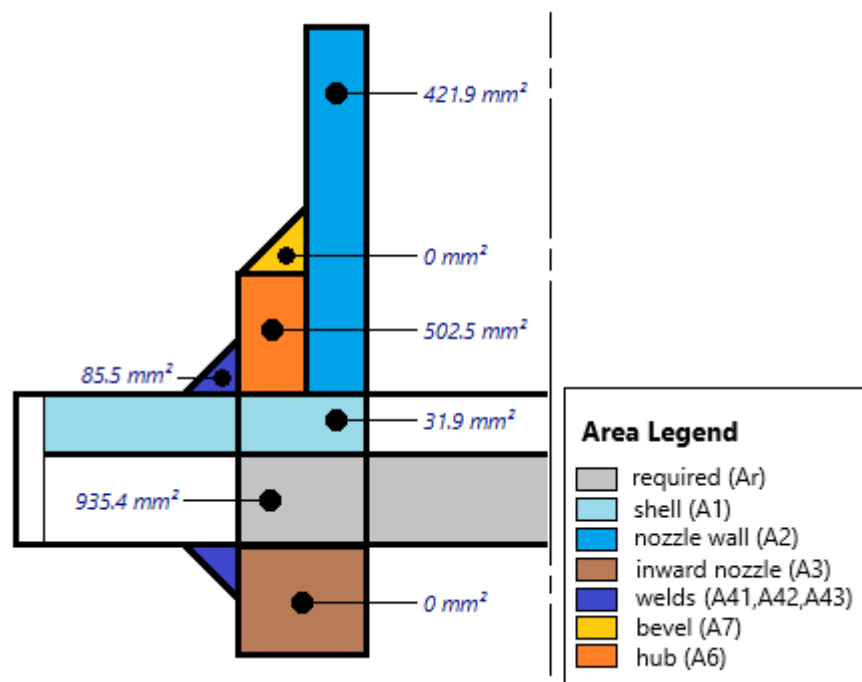
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	144.8618	mm
Parallel to Vessel Wall, opening length	d	72.4309	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	31.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	935.382	NA	NA
Area in Shell	A1	31.928	NA	NA
Area in Nozzle Wall	A2	421.931	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.507	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	502.469	NA	NA
TOTAL AREA AVAILABLE	Atot	1041.835	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (72.4 \cdot 12 \cdot 1 + 2 \cdot 18.5 \cdot 12 \cdot 1 \cdot (1 - 0.86))$
 $= 935.382 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 72.4 (1 \cdot 12.5 - 1 \cdot 12) - 2 \cdot 18.5$
 $(1 \cdot 12.5 - 1 \cdot 12) \cdot (1 - 0.86)$
 $= 31.928 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 31.3) (8.23 - 0.34) 0.86$
 $= 421.931 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 85.507 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(31.3, 200, 24)) \cdot (20 - 9.73) \cdot 0.86$$

$$= 502.469 \text{ mm}^2 \text{ Includes Bevel Area of } 81.139 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 1.8394 \text{ mm}$
 Wall Thickness per UG16(b), $tr_{16b} = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb_1 = 13.5239 \text{ mm}$
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 13.5239 \text{ mm}$
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb_3 = 6.3000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[6.3, \max(13.5, 3)]$$

$$= 6.3000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1.84, 6.3)$$

$$= 6.3000 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	30.5,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	264.5 MPa	Passed
Occasional	:	2.2,	Allowable	:	156.9 MPa	Passed
Shear	:	22.3,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $1.10/8.52 = 0.129$

Weld Size Calculations, Description: A23"

Intermediate Calc. for nozzle/shell Welds	Tmin	12.5000 mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

$$\begin{aligned} \text{Weld Load [W]:} \\ &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (935 - 31.9 + 2 \cdot 18.5 \cdot 0.86 \cdot (1 \cdot 12.5 - 12)) \cdot 138) \\ &= 126741.94 \text{ N} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

$$\begin{aligned} \text{Weld Load [W1]:} \\ &= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (924 + 85.5 - 0 \cdot 0.86) \cdot 138 \\ &= 139352.72 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W2]:} \\ &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (924 + 0 + 85.5 + (395)) \cdot 138 \\ &= 193922.09 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W3]:} \\ &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (924 + 0 + 85.5 + 0 + (395)) \cdot 138 \\ &= 193922.09 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

$$\begin{aligned} \text{Shear, Outward Nozzle Weld [Sonw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 109 \cdot 10 \cdot 0.49 \cdot 118 \\ &= 99379. \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Shear, Nozzle Wall [Snw]:} \\ &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 45.5) \cdot (20 - 1.5) \cdot 0.7 \cdot 118 \\ &= 218242. \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Tension, Shell Groove Weld [Tngw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 109 \cdot (12 - 0) \cdot 0.74 \cdot 138 \\ &= 210624. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

$$\begin{aligned} \text{Failure Path [Path 1-1]:} \\ &= (\text{SONW} + \text{SNW}) = (99379 + 218242) = 317621 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 2-2]:} \\ &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (99379 + 0 + 210624 + 0) = 310002 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 3-3]:} \\ &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ &= (99379 + 210624 + 0) = 310002 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 317620 N	, must exceed W = 126741 N	or W1 = 139352 N
Path 2-2 = 310002 N	, must exceed W = 126741 N	or W2 = 193922 N
Path 3-3 = 310002 N	, must exceed W = 126741 N	or W3 = 193922 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.13 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.



The Drop for this Nozzle is : 0.9983 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.9983 mm

Input Echo, WRC297 Item 1, Description: A23" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3000.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	69.431	mm
Nozzle Thickness	20.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc 118.00	MPa
Nozzle Hot S.I. Allowable	SNmh 118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	1.10	MPa
Radial Load	P	-4799.79	N
Circumferential Shear	Vc	3599.84	N
Longitudinal Shear	Vl	4799.79	N
Circumferential Moment	Mc	720253.69	N-mm
Longitudinal Moment	Ml	936329.88	N-mm
Torsional Moment	Mt	0.108E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.472) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 109.431 + 2 \cdot 1.65 \cdot \sqrt{(1507.75(14.0 - 1.5))}$
 $= 562.468 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3015.500	mm
Nozzle Outside Diameter	(d) =	109.431	mm
Vessel Thickness used	(T) =	12.500	mm
Nozzle Thickness used	(t) =	18.500	mm
T / t	=	0.676	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.564$$

Nr/P	=	0.145
Mr/P	=	0.292



M0/P = 0.092
 N0/P = 0.166
 MrD/Mc = 0.305
 NrDL/Mc = 0.103
 M0d/Mc = 0.105
 N0DL/Mc = -0.050
 MrD/Ml = 0.257
 NrDL/Ml = 0.105
 M0D/Ml = 0.089
 N0DL/Ml = 0.044

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-29	29	29	-29
Normal Pressure Stress	131	132	131	132
Outplane Stress Summary	121	147	185	93

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-84	84	84	-84
Inplane Pressure Stress	65	65	65	65
Inplane Stress Summary	33	95	214	-62

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	122	148	214	156

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-26	26	26	-26
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



Normal Pressure Stress	65	65	65	65
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Outplane Stress Summary	63	82	112	25
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-76	76	76	-76
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	131	132	131	132

Inplane Stress Summary	108	155	271	10
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	6	6
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	108	156	271	27
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3

Hoop Stress Summary	4	6	9	10
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-8	-8	8	8
Axial Bending (ML)	-30	30	30	-30
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-19	5	60	-38
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3

Shear Stress Summary	4	4	1	1
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	25	9	60	49
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3

Hoop Stress Summary	9	10	5	6
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-29	29	29	-29
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-15	5	56	-38
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	3	3	3	3

Shear Stress Summary	1	1	4	4
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	25	11	56	46
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	131	132	131	132	131	132	131	132
Circ. Pl (SUS)	3	3	7	7	0	0	8	8
Circ. Q (SUS)	-13	13	45	-45	-23	23	129	-129



Long. Pm (SUS)	65	65	65	65	65	65	65	65
Long. Pl (SUS)	-1	-1	9	9	7	7	3	3
Long. Q (SUS)	-31	31	137	-137	-10	10	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	131.0	132.0	131.0	132.0	131.0	132.0	131.0	132.0
Pm+Pl (SUS)	134.0	135.0	138.0	139.0	131.1	132.1	139.1	140.1
Pm+Pl+Q (Total)	121.3	148.5	211.3	157.1	108.1	155.1	268.2	28.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	132.00	138.00	Passed
Pm+Pl (SUS)	140.06	207.00	Passed
Pm+Pl+Q (TOTAL)	268.23	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	3	2	3	2	3	2	3
Circ. Pl (SUS)	3	3	7	7	7	7	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-8	-8	8	8	-6	-6	6	6
Long. Q (SUS)	-12	12	48	-48	-11	11	47	-47
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	2.0	3.0	2.0	3.0	2.0	3.0	2.0	3.0
Pm+Pl (SUS)	12.2	13.2	10.0	10.6	14.1	15.1	7.4	7.6
Pm+Pl+Q (Total)	25.3	9.5	57.1	49.2	25.3	10.8	54.3	46.7

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	118.00	Passed
Pm+Pl (SUS)	15.13	177.00	Passed
Pm+Pl+Q (TOTAL)	57.08	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input, Nozzle Desc: A32"

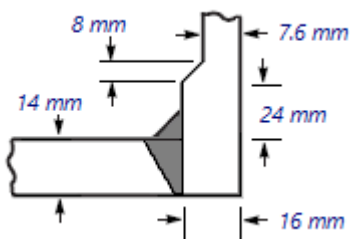
From: 60

Pressure for Reinforcement Calculations	P	1.1031	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6050.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	24.0000	mm
Height of Beveled Transition	L'	8.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		16.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: A32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1502 / (138 \cdot 1 - 0.6 \cdot 1.1)$
 $= 12.0606 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 1.1)$
 $= 0.2258 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 8.35 / 0.5773$
 $= 14.4706 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	96.0684	mm
Parallel to Vessel Wall, opening length	d	48.0342	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	29.8341	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: A32".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7258	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 13.5606	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.5606	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(13.6, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.73, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	39.6,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	255.4 MPa	Passed
Occasional	:	1.9,	Allowable	:	156.9 MPa	Passed
Shear	:	30.2,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.



Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: A32"

Intermediate Calc. for nozzle/shell Welds Tmin 12.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4946 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.4947 mm

Input Echo, WRC297 Item 1, Description: A32" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3000.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	45.034	mm
Nozzle Thickness	16.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	1.10	MPa
Radial Load	P	-3199.86	N
Circumferential Shear	Vc	2399.89	N
Longitudinal Shear	Vl	3199.86	N
Circumferential Moment	Mc	320112.56	N-mm
Longitudinal Moment	Ml	416146.78	N-mm
Torsional Moment	Mt	480169.53	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.815) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 77.034 + 2 \cdot 1.65 \cdot \sqrt{1507.75(14.0 - 1.5)}$$

$$= 530.071 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 3015.500 mm
 Nozzle Outside Diameter (d) = 77.034 mm
 Vessel Thickness used (T) = 12.500 mm
 Nozzle Thickness used (t) = 14.500 mm
 T / t = 0.862
 d / T = 10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 0.397$$

Nr/P = 0.181
 Mr/P = 0.327
 M0/P = 0.110
 N0/P = 0.188
 MrD/Mc = 0.298
 NrDL/Mc = 0.124
 M0d/Mc = 0.114
 N0DL/Mc = -0.092
 MrD/Ml = 0.277
 NrDL/Ml = 0.125
 M0D/Ml = 0.104
 N0DL/Ml = -0.018

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-21	21	21	-21
Normal Pressure Stress	131	133	131	133
Outplane Stress Summary	128	145	170	101

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	40	-40	40	-40
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-57	57	57	-57
Inplane Pressure Stress	65	65	65	65
Inplane Stress Summary	48	82	171	-23

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4



Shear Stress Summary	5	5	2	2
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	128	146	173	124
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65	65	65	65

Outplane Stress Summary	67	76	99	35
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	40	-40	40	-40
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-47	47	47	-47
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	131	133	131	133

Inplane Stress Summary	124	140	226	52
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	6	6
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	125	140	226	54
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3



Hoop Stress Summary	6	7	4	6
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	21	-21	21	-21
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-33	33	33	-33
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-20	2	67	-41
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3

Shear Stress Summary	4	4	2	2
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	28	11	67	48
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3

Hoop Stress Summary	7	9	3	4
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	21	-21	21	-21
Axial Membrane (Mc)	-8	-8	8	8
Axial Bending (Mc)	-28	28	28	-28
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-12	0	59	-39
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1

Shear due to Torsion 3 3 3 3

Shear Stress Summary 1 1 5 5

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	21	9	60	44

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	131	133	131	133	131	133	131	133
Circ. Pl (SUS)	3	3	3	3	0	0	6	6
Circ. Q (SUS)	-8	8	34	-34	-7	7	87	-87
Long. Pm (SUS)	65	65	65	65	65	65	65	65
Long. Pl (SUS)	-1	-1	7	7	5	5	1	1
Long. Q (SUS)	-17	17	97	-97	-5	5	31	-31
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	131.0	133.0	131.0	133.0	131.0	133.0	131.0	133.0
Pm+Pl (SUS)	134.0	136.0	134.0	136.0	131.1	133.1	137.1	139.1
Pm+Pl+Q (Total)	126.3	144.4	171.5	127.1	124.1	140.1	224.3	53.9

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	133.00	138.00	Passed
Pm+Pl (SUS)	139.05	207.00	Passed
Pm+Pl+Q (TOTAL)	224.28	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	3	1	3	1	3	1	3
Circ. Pl (SUS)	3	3	3	3	5	5	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-9	-9	11	11	-7	-7	9	9
Long. Q (SUS)	-12	12	54	-54	-7	7	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	1.0	3.0	1.0	3.0	1.0	3.0	1.0	3.0
Pm+Pl (SUS)	13.2	15.1	11.1	11.2	13.2	15.1	9.1	9.2



Pm+Pl+Q (Total)	26.2	8.8	65.1	49.2	20.4	8.9	58.3	44.7
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Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	118.00	Passed
Pm+Pl (SUS)	15.13	177.00	Passed
Pm+Pl+Q (TOTAL)	65.07	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: A118"

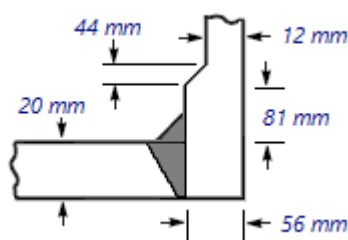
From: 70

Pressure for Reinforcement Calculations	P	1.0936	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		8075.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	81.0000	mm
Height of Beveled Transition	L'	44.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		56.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: A118"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 18.000 in.
 Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.09 \cdot 1502 / (138 \cdot 1 - 0.6 \cdot 1.09)$
 $= 11.9562 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.09 \cdot 230 / (118 \cdot 1 - 0.6 \cdot 1.09)$
 $= 2.1445 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 44 / 0.5773$
 $= 76.2103 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

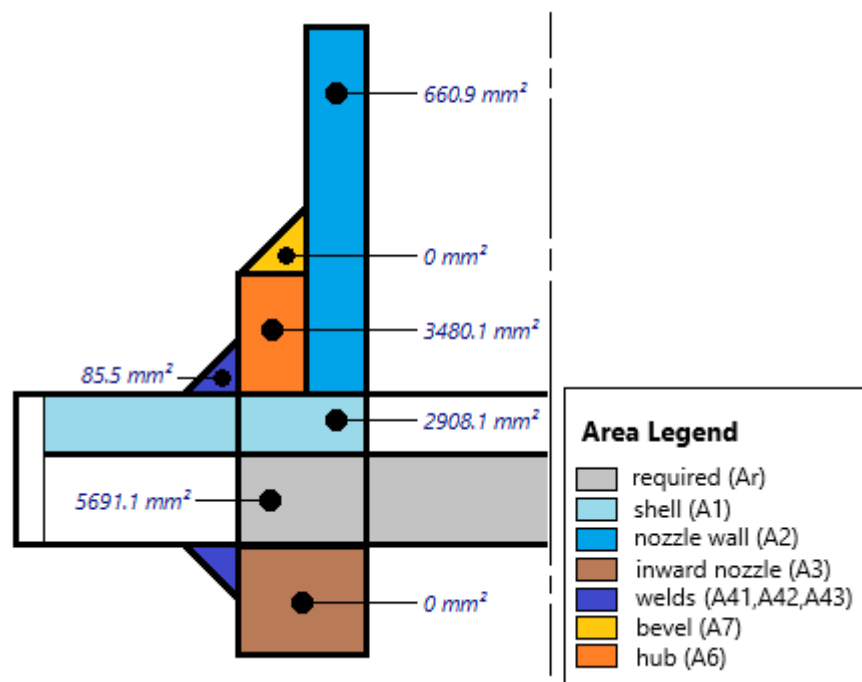
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	920.4000	mm
Parallel to Vessel Wall, opening length	d	460.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5691.124	NA	NA
Area in Shell	A1	2908.076	NA	NA
Area in Nozzle Wall	A2	660.869	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.507	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	3480.145	NA	NA
TOTAL AREA AVAILABLE	Atot	7134.597	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (460 \cdot 12 \cdot 1 + 2 \cdot 54.5 \cdot 12 \cdot 1 \cdot (1 - 0.86)) \\
 &= 5691.124 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 460(1 \cdot 18.5 - 1 \cdot 12) - 2 \cdot 54.5 \\
 &\quad (1 \cdot 18.5 - 1 \cdot 12) \cdot (1 - 0.86) \\
 &= 2908.076 \text{ mm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 46.2)(10.5 - 2.14)0.86 \\
 &= 660.869 \text{ mm}^2
 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\
 &= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 \\
 &= 85.507 \text{ mm}^2
 \end{aligned}$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(46.2, 250, 81)) \cdot (56 - 12) \cdot 0.86$$

$$= 3480.145 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.6445 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 13.4562 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 13.4562 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 9.8312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.83, \max(13.5, 3)]$$

$$= 9.8312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.64, 9.83)$$

$$= 9.8312 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	33.6,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	261.4 MPa	Passed
Occasional	:	11.7,	Allowable	:	156.9 MPa	Passed
Shear	:	15.9,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-47 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $1.09/1.63 = 0.671$

Weld Size Calculations, Description: A118"

Intermediate Calc. for nozzle/shell Welds	Tmin	18.5000 mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (5691 - 2908 + 2 \cdot 54.5 \cdot 0.86 \cdot (1 \cdot 18.5 - 12)) \cdot 138)$$

$$= 468178.47 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (4141 + 85.5 - 0 \cdot 0.86) \cdot 138$$

$$= 583199.75 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (4141 + 0 + 85.5 + (1724)) \cdot 138$$

$$= 821122.25 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (4141 + 0 + 85.5 + 0 + (1724)) \cdot 138$$

$$= 821122.25 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 569 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 516914. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 257) \cdot (56 - 1.5) \cdot 0.7 \cdot 118$$

$$= 3639199. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 569 \cdot (18 - 0) \cdot 0.74 \cdot 138$$

$$= 1643324. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (516914 + 3639199) = 4156113 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (516914 + 0 + 1643324 + 0) = 2160238 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (516914 + 1643324 + 0) = 2160238 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 4156113 N	, must exceed W = 468178 N	or W1 = 583199 N
Path 2-2 = 2160238 N	, must exceed W = 468178 N	or W2 = 821122 N
Path 3-3 = 2160238 N	, must exceed W = 468178 N	or W3 = 821122 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.23 MPa

The Drop for this Nozzle is : 27.2465 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 297.2465 mm

Input Echo, WRC297 Item 1, Description: A118" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3000.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	457.200	mm
Nozzle Thickness	56.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	1.09	MPa
Radial Load	P	-28798.71	N
Circumferential Shear	Vc	21599.00	N
Longitudinal Shear	Vl	28798.71	N
Circumferential Moment	Mc	0.259E+08	N-mm
Longitudinal Moment	Ml	0.337E+08	N-mm
Torsional Moment	Mt	0.389E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 569.2 + 2 \cdot 1.65 \cdot \sqrt{(1510.75(20.0 - 1.5))}$
 $= 1120.891 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3021.500	mm
Nozzle Outside Diameter	(d) =	569.200	mm
Vessel Thickness used	(T) =	18.500	mm
Nozzle Thickness used	(t) =	54.500	mm
T / t	=	0.339	
d / t	=	10.444	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.408$$

Nr/P	=	0.063
Mr/P	=	0.125
M0/P	=	0.038
N0/P	=	0.081
MrD/Mc	=	0.278
NrDL/Mc	=	0.095



M0d/Mc = 0.085
 N0DL/Mc = 0.081
 MrD/Ml = 0.101
 NrDL/Ml = 0.068
 M0D/Ml = 0.030
 N0DL/Ml = 0.102

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-17	-17	17	17
Outplane Bending (ML)	-30	30	30	-30
Normal Pressure Stress	88	89	88	89
Outplane Stress Summary	65	90	162	63

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-104	104	104	-104
Inplane Pressure Stress	44	44	44	44
Inplane Stress Summary	-3	78	228	-106

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	69	92	228	170

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-67	67	67	-67
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	44	44	44	44
Outplane Stress Summary	-8	88	148	-24

**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-221	221	221	-221
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	88	89	88	89
Inplane Stress Summary	-77	240	390	-177

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	5	5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	77	240	390	177

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-17	-17	17	17
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	6	4	6
Hoop Stress Summary	-5	-4	29	30

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-7	7	7	-7
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-2	2	19	-7

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1



Shear Stress Summary	1	1	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	6	7	29	38
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	6	4	6

Hoop Stress Summary	0	2	22	23
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-21	21	21	-21
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-15	16	31	-21
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	1	1
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	16	16	32	45
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	88	89	88	89	88	89	88	89
Circ. Pl (SUS)	-11	-11	23	23	-7	-7	17	17
Circ. Q (SUS)	-11	11	49	-49	-158	158	284	-284
Long. Pm (SUS)	44	44	44	44	44	44	44	44
Long. Pl (SUS)	-6	-6	16	16	-4	-4	16	16
Long. Q (SUS)	-41	41	167	-167	-48	48	86	-86



Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	88.0	89.0	88.0	89.0	88.0	89.0	88.0	89.0
Pm+Pl (SUS)	77.0	78.0	111.0	112.0	81.0	82.0	105.0	106.0
Pm+Pl+Q (Total)	69.7	91.1	227.1	170.1	77.1	240.1	389.1	178.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	89.00	138.00	Passed
Pm+Pl (SUS)	112.02	207.00	Passed
Pm+Pl+Q (TOTAL)	389.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	6	4	6	4	6	4	6
Circ. Pl (SUS)	-11	-11	23	23	-4	-4	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-3	-3	3	3	-2	-2	2	2
Long. Q (SUS)	-2	2	12	-12	-16	16	26	-26
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	4.0	6.0	4.0	6.0	4.0	6.0	4.0	6.0
Pm+Pl (SUS)	7.0	5.0	27.0	29.0	0.0	2.0	20.0	22.0
Pm+Pl+Q (Total)	7.2	6.3	27.1	36.1	16.1	16.1	30.1	44.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	118.00	Passed
Pm+Pl (SUS)	29.00	177.00	Passed
Pm+Pl+Q (TOTAL)	44.05	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

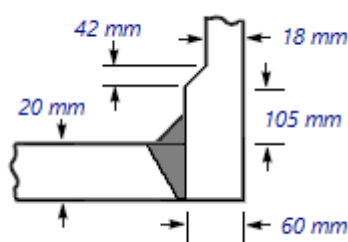
Input, Nozzle Desc: M124"
From: 70

Pressure for Reinforcement Calculations	P	1.0936	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		8250.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	105.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).


Reinforcement CALCULATION, Description: M124"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.000 in.
 Actual Thickness Used in Calculation 0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.09 \cdot 1502 / (138 \cdot 1 - 0.6 \cdot 1.09)$
 $= 11.9562 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.09 \cdot 305 / (118 \cdot 1 + 0.4 \cdot 1.09)$
 $= 2.8145 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 42 / 0.5773$
 $= 72.7462 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

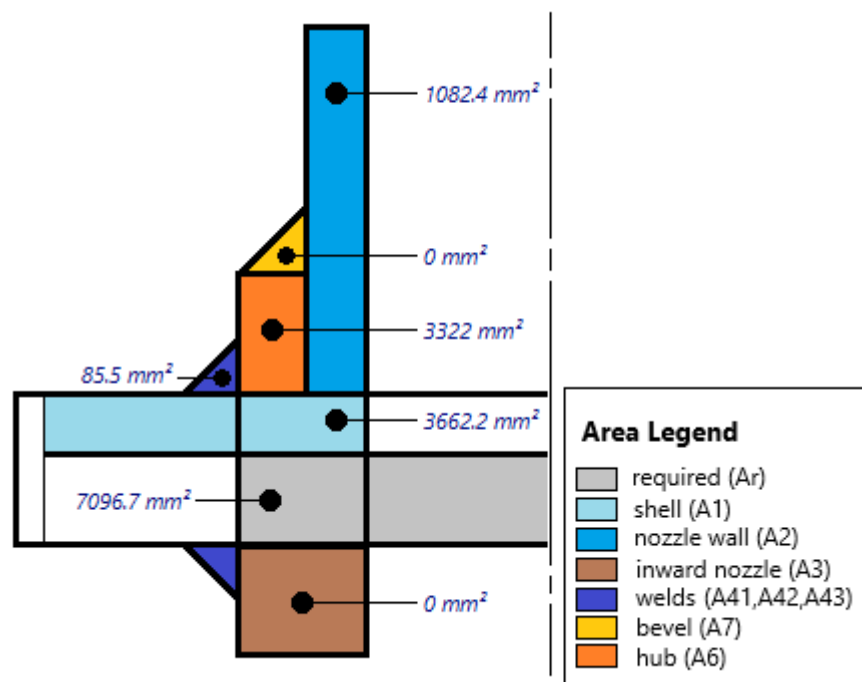
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1153.2001	mm
Parallel to Vessel Wall, opening length	d	576.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7096.691	NA	NA
Area in Shell	A1	3662.186	NA	NA
Area in Nozzle Wall	A2	1082.443	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.507	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	3321.957	NA	NA
TOTAL AREA AVAILABLE	Atot	8152.092	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (577 \cdot 12 \cdot 1 + 2 \cdot 58.5 \cdot 12 \cdot 1 \cdot (1 - 0.86))$$

$$= 7096.691 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 577(1 \cdot 18.5 - 1 \cdot 12) - 2 \cdot 58.5(1 \cdot 18.5 - 1 \cdot 12) \cdot (1 - 0.86)$$

$$= 3662.186 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 46.2)(16.5 - 2.81)0.86$$

$$= 1082.443 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$$

$$= 85.507 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(46.2, 300, 105)) \cdot (60 - 18) \cdot 0.86$$

$$= 3321.957 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.3145 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= t_a$
 $= 4.3145 \text{ mm}$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-29 \text{ }^\circ\text{C}$

Calculated Minimum Design Metal Temperature $-104 \text{ }^\circ\text{C}$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-29 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-29 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification $-29 \text{ }^\circ\text{C}$

Flange MDMT with Temp reduction per UCS-66(i)(2) $-47 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

Design Pressure/Ambient Rating = $1.09/1.63 = 0.671$

Weld Size Calculations, Description: M124"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 18.5000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (7097 - 3662 + 2 \cdot 58.5 \cdot 0.86 \cdot (1 \cdot 18.5 - 12)) \cdot 138)$$

$$= 564246.94 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - C_{an} / .707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (4404 + 85.5 - 0 \cdot 0.86) \cdot 138$$

$$= 619543.19 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (4404 + 0 + 85.5 + (1851)) \cdot 138$$

= 874927.94 N

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S$$

$$= (4404 + 0 + 85.5 + 0 + (1851)) \cdot 138$$

$$= 874927.94 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 553603. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 297) \cdot (60 - 1.5) \cdot 0.7 \cdot 118$$

$$= 4501310. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{g n v i} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 610 \cdot (12 - 0) \cdot 0.74 \cdot 138$$

$$= 1173308. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (553603 + 4501310) = 5054913 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (553603 + 0 + 1173308 + 0) = 1726910 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (553603 + 1173308 + 0) = 1726910 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 5054913 N	, must exceed W = 564246 N	or W1 = 619543 N
Path 2-2 = 1726910 N	, must exceed W = 564246 N	or W2 = 874927 N
Path 3-3 = 1726910 N	, must exceed W = 564246 N	or W3 = 874927 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.18 MPa

The Drop for this Nozzle is : 31.2941 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 351.2942 mm

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 205.72 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 576.60 mm

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Input, Nozzle Desc: L12"

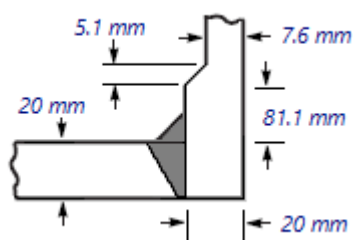
From: 70

Pressure for Reinforcement Calculations	P	1.0936	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7500.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		315.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	81.1239	mm
Height of Beveled Transition	L'	5.1239	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.09 \cdot 1502 / (138 \cdot 1 - 0.6 \cdot 1.09)$
 $= 11.9562 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.09 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 1.09)$
 $= 0.2238 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 18.5$, less than the hub height, use (e-2)
 $= 46.2500 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	122.0342	mm
Parallel to Vessel Wall	Rn+tn+t	61.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Hub Height was ≥ 2.5 times Hub Thickness, using sketch UG-40 (e-2).

*Taking a UG-36(c)(3)(a) exemption for nozzle: L12".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7238	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 13.4562	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.4562	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(13.5, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.72, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Weld Size Calculations, Description: L12"



Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6027 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.6027 mm

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Input, Nozzle Desc: L32"

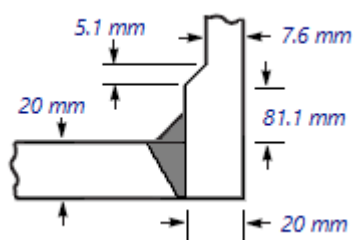
From: 70

Pressure for Reinforcement Calculations	P	1.0936	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7500.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	81.1239	mm
Height of Beveled Transition	L'	5.1239	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.09 \cdot 1502 / (138 \cdot 1 - 0.6 \cdot 1.09)$
 $= 11.9562 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.09 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 1.09)$
 $= 0.2238 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 18.5$, less than the hub height, use (e-2)
 $= 46.2500 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	122.0342	mm
Parallel to Vessel Wall	Rn+tn+t	61.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Hub Height was ≥ 2.5 times Hub Thickness, using sketch UG-40 (e-2).

*Taking a UG-36(c)(3)(a) exemption for nozzle: L32".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7238	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 13.4562	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.4562	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(13.5, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.72, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Weld Size Calculations, Description: L32"



Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6027 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.6027 mm

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Input, Nozzle Desc: L52'

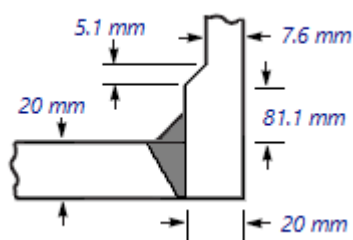
From: 70

Pressure for Reinforcement Calculations	P	1.0936	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7500.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	81.1239	mm
Height of Beveled Transition	L'	5.1239	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L52'

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.09 \cdot 1502 / (138 \cdot 1 - 0.6 \cdot 1.09)$
 $= 11.9562 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.09 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 1.09)$
 $= 0.2238 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 18.5$, less than the hub height, use (e-2)
 $= 46.2500 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	122.0342	mm
Parallel to Vessel Wall	Rn+tn+t	61.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Hub Height was ≥ 2.5 times Hub Thickness, using sketch UG-40 (e-2).

*Taking a UG-36(c)(3)(a) exemption for nozzle: L52'.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7238	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 13.4562	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.4562	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(13.5, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.72, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Weld Size Calculations, Description: L52'



Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6027 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.6027 mm

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Input, Nozzle Desc: L72"

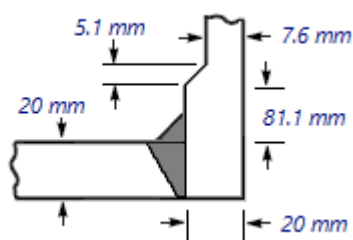
From: 70

Pressure for Reinforcement Calculations	P	1.0936	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7500.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	81.1239	mm
Height of Beveled Transition	L'	5.1239	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L72"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.773 in.
 Actual Thickness Used in Calculation 0.301 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.09 \cdot 1502 / (138 \cdot 1 - 0.6 \cdot 1.09)$
 $= 11.9562 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.09 \cdot 24 / (118 \cdot 1 - 0.6 \cdot 1.09)$
 $= 0.2238 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 18.5$, less than the hub height, use (e-2)
 $= 46.2500 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	122.0342	mm
Parallel to Vessel Wall	Rn+tn+t	61.0171	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.2500	mm

Hub Height was ≥ 2.5 times Hub Thickness, using sketch UG-40 (e-2).

*Taking a UG-36(c)(3)(a) exemption for nozzle: L72".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7238	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 13.4562	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.4562	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(13.5, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.72, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 7.6454 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Weld Size Calculations, Description: L72"



Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6027 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.6027 mm

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Input, Nozzle Desc: B118"

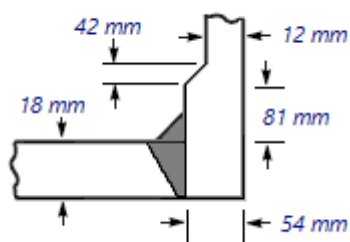
From: 80

Pressure for Reinforcement Calculations	P	1.0936	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	3000.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	18.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	81.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		54.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: B118"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 18.000 in.
 Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.09 \cdot 3003 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.09)$
 $= 10.7077 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.09 \cdot 230 / (118 \cdot 1 - 0.6 \cdot 1.09)$
 $= 2.1445 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 42 / 0.5773$
 $= 72.7462 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

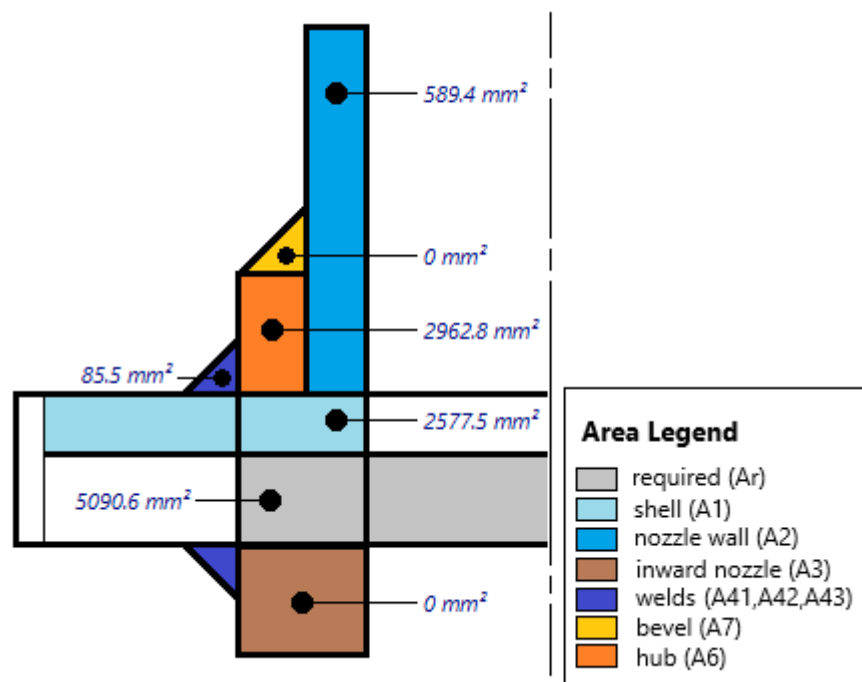
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	920.4000	mm
Parallel to Vessel Wall, opening length	d	460.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5090.628	NA	NA
Area in Shell	A1	2577.472	NA	NA
Area in Nozzle Wall	A2	589.423	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.507	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	2962.826	NA	NA
TOTAL AREA AVAILABLE	Atot	6215.229	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (460 \cdot 10.7 \cdot 1 + 2 \cdot 52.5 \cdot 10.7 \cdot 1 \cdot (1 - 0.86))$$

$$= 5090.628 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 460(1 \cdot 16.5 - 1 \cdot 10.7) - 2 \cdot 52.5$$

$$(1 \cdot 16.5 - 1 \cdot 10.7) \cdot (1 - 0.86)$$

$$= 2577.472 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 41.2)(10.5 - 2.14)0.86$$

$$= 589.423 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$$

$$= 85.507 \text{ mm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(41.2, 250, 81)) \cdot (54 - 12) \cdot 0.86$$

$$= 2962.826 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6445 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.3930 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.3930 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 9.8312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.83, \max(13.4, 3)]$$

$$= 9.8312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.64, 9.83)$$

$$= 9.8312 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 30.3,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 264.7 MPa	Passed
Occasional	: 11.7,	Allowable	: 156.9 MPa	Passed
Shear	: 11.9,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-29 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-47 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 1.09/1.63 = 0.671$$

Weld Size Calculations, Description: B118"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

$$\begin{aligned} \text{Weld Load [W]:} \\ &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (5091 - 2577 + 2 \cdot 52.5 \cdot 0.86 \cdot \\ &\quad (1 \cdot 16.5 - 10.7)) \cdot 138) \\ &= 418538.91 \text{ N} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

$$\begin{aligned} \text{Weld Load [W1]:} \\ &= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (3552 + 85.5 - 0 \cdot 0.86) \cdot 138 \\ &= 501958.66 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W2]:} \\ &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (3552 + 0 + 85.5 + (1481)) \cdot 138 \\ &= 706372.62 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W3]:} \\ &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (3552 + 0 + 85.5 + 0 + (1481)) \cdot 138 \\ &= 706372.62 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

$$\begin{aligned} \text{Shear, Outward Nozzle Weld [Sonw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 565 \cdot 10 \cdot 0.49 \cdot 118 \\ &= 513281. \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Shear, Nozzle Wall [Snw]:} \\ &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 256) \cdot (54 - 1.5) \cdot 0.7 \cdot 118 \\ &= 3492028. \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Tension, Shell Groove Weld [Tngw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 565 \cdot (12 - 0) \cdot 0.74 \cdot 138 \\ &= 1087851. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

$$\begin{aligned} \text{Failure Path [Path 1-1]:} \\ &= (\text{SONW} + \text{SNW}) = (513281 + 3492028) = 4005309 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 2-2]:} \\ &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (513281 + 0 + 1087850 + 0) = 1601132 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 3-3]:} \\ &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ &= (513281 + 1087850 + 0) = 1601132 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 4005309 N	, must exceed W = 418538 N	or W1 = 501958 N
Path 2-2 = 1601132 N	, must exceed W = 418538 N	or W2 = 706372 N
Path 3-3 = 1601132 N	, must exceed W = 418538 N	or W3 = 706372 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.22 MPa

The Drop for this Nozzle is : 14.7508 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 282.7509 mm

Input Echo, WRC107/537 Item 1, Description: B118" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	5400.000	mm
Vessel Thickness	Tv	18.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	457.200	mm
Nozzle Thickness	Tn	54.000	mm
Nozzle Material		SA-350 LF1	
Nozzle UNS Number		K03009	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Design Internal Pressure	Dp	1.094	MPa
Include Pressure Thrust		No	

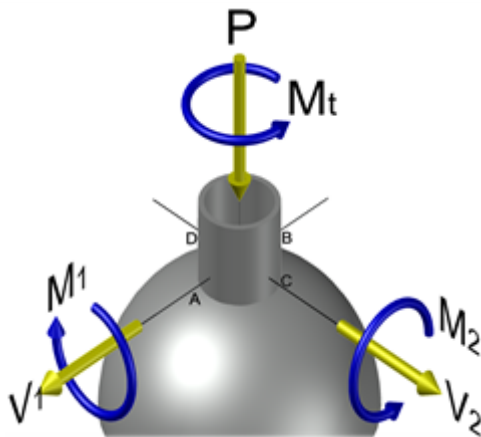
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-21600.0	N
Longitudinal Shear (SUS)	(V1) V1	21600.0	N
Circumferential Shear (SUS)	(Vc) V2	16200.0	N
Circumferential Moment (SUS)	(Mc) M1	25271940.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	25271940.0	N-mm
Torsional Moment (SUS)	Mt	29160044.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 565.2 + 2 \cdot 1.65 \cdot \sqrt{(2709.75 (18.0 - 1.5))}$$

$$= 1262.983 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	-21600.0	N
Circumferential Shear	(VC)	V2	16200.0	N
Longitudinal Shear	(VL)	V1	21600.0	N
Circumferential Moment	(MC)	M1	25271940.0	N-mm
Longitudinal Moment	(ML)	M2	25271940.0	N-mm
Torsional Moment		MT	29160044.0	N-mm

Dimensionless Param: U = 1.34 TAU = 5.00 (4.88) RHO = 0.31

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.02967	(A,B,C,D)
$M(x) / P$	SP 1	0.02977	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03025	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.04456	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03025	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.04456	(A,B,C,D)
$N(y) * T / P$	SP 1	0.04373	(A,B,C,D)
$M(y) / P$	SP 1	0.00821	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04099	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01335	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04099	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01335	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4
Rad. Bend.	P	14.2	-14.2	14.2	-14.2	14.2	-14.2	14.2	-14.2
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-13.3	-13.3	13.3	13.3
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-117.3	117.3	117.3	-117.3
Rad. Memb.	ML	-13.3	-13.3	13.3	13.3	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-117.3	117.3	117.3	-117.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-114.1	92.2	147.1	-115.9	-114.1	92.2	147.1	-115.9



Tang. Memb. P	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
Tang. Bend. P	3.9	-3.9	3.9	-3.9	3.9	-3.9	3.9	-3.9
Tang. Memb. MC	0.0	0.0	0.0	0.0	-18.0	-18.0	18.0	18.0
Tang. Bend. MC	0.0	0.0	0.0	0.0	-35.2	35.2	35.2	-35.2
Tang. Memb. ML	-18.0	-18.0	18.0	18.0	0.0	0.0	0.0	0.0
Tang. Bend. ML	-35.2	35.2	35.2	-35.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-45.8	16.7	60.5	-17.6	-45.8	16.7	60.5	-17.6
Shear VC	1.1	1.1	-1.1	-1.1	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Shear MT	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
Tot. Shear	4.6	4.6	2.4	2.4	2.0	2.0	5.0	5.0
Str. Int.	114.4	92.5	147.2	115.9	114.1	92.3	147.4	116.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	89.5	89.5	89.5	89.5	89.5	89.5	89.5	89.5
Rad.	Pl (SUS)	-10.9	-10.9	15.6	15.6	-10.9	-10.9	15.6	15.6
Rad.	Q (SUS)	-103.2	103.2	131.5	-131.5	-103.2	103.2	131.5	-131.5
Long.	Pm (SUS)	89.5	89.5	89.5	89.5	89.5	89.5	89.5	89.5
Long.	Pl (SUS)	-14.5	-14.5	21.5	21.5	-14.5	-14.5	21.5	21.5
Long.	Q (SUS)	-31.2	31.2	39.1	-39.1	-31.2	31.2	39.1	-39.1
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.1	1.1	-1.1	-1.1	-1.5	-1.5	1.5	1.5
Shear	Q (SUS)	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
Pm (SUS)		89.5	89.5	89.5	89.5	89.5	89.5	89.5	89.5
Pm+Pl (SUS)		78.9	78.9	111.2	111.2	79.1	79.1	111.3	111.3
Pm+Pl+Q (Total)		68.9	182.1	236.7	98.4	68.4	181.8	237.0	98.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	89.53	138.00	Passed
Pm+Pl (SUS)	111.34	207.00	Passed
Pm+Pl+Q (TOTAL)	236.96	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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